PHYSICS MODULE 8

From the Universe to the Atom

PHYSICS MODULE 8: FROM THE UNIVERSE TO THE ATOM

ORIGINS OF THE ELEMENTS

Inquiry question: What evidence is there for the origins of the elements?

- investigate the processes that led to the transformation of radiation into matter that followed the 'Big Bang'
- investigate the evidence that led to the discovery of the expansion of the Universe by Hubble (ACSPH138)
- analyse and apply Einstein's description of the equivalence of energy and mass and relate this to the nuclear reactions that occur in stars (ACSPH031)
- account for the production of emission and absorption spectra and compare these with a continuous black body spectrum (ACSPH137)
- investigate the key features of stellar spectra and describe how these are used to classify stars
- investigate the Hertzsprung–Russell diagram and how it can be used to determine the following about a star:
  - characteristics and evolutionary stage
  - surface temperature
  - colour
  - luminosity
- investigate the types of nucleosynthesis reactions involved in Main Sequence and Post-Main Sequence stars, including but not limited to:
  - proton–proton chain
  - CNO (carbon–nitrogen–oxygen) cycle

STRUCTURE OF THE ATOM

Inquiry question: How is it known that atoms are made up of protons, neutrons and electrons?

- investigate, assess and model the experimental evidence supporting the existence and properties of the electron, including:
  - early experiments examining the nature of cathode rays
  - Thomson's charge-to-mass experiment
  - Millikan's oil drop experiment (ACSPH026)
- investigate, assess and model the experimental evidence supporting the nuclear model of the atom, including:
  - the Geiger–Marsden experiment
  - Rutherford's atomic model
  - Chadwick's discovery of the neutron (ACSPH026)

QUANTUM MECHANICAL NATURE OF THE ATOM

Inquiry question: How is it known that classical physics cannot explain the properties of the atom?

- assess the limitations of the Rutherford and Bohr atomic models
- investigate the line emission spectra to examine the Balmer series in hydrogen (ACSPH138)
- relate qualitatively and quantitatively the quantised energy levels of the hydrogen atom and the law of conservation of energy to the line emission spectrum of hydrogen using:
  - $E = hf$
  - $E = \frac{hc}{\lambda}$
  - $\frac{1}{\lambda} = R\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right]$ (ACSPH136)
- investigate de Broglie's matter waves, and the experimental evidence that developed the following formula:
  - $\lambda = \frac{h}{mv}$ (ACSPH140)
- analyse the contribution of Schrödinger to the current model of the atom

PROPERTIES OF THE NUCLEUS

Inquiry question: How can the energy of the atomic nucleus be harnessed?

- analyse the spontaneous decay of unstable nuclei, and the properties of the alpha, beta and gamma radiation emitted (ACSPH028, ACSPH030)
- examine the model of half-life in radioactive decay and make quantitative predictions about the activity or amount of a radioactive sample using the following relationships:
  - $N_t = N_0 e^{-\lambda t}$
  - $\lambda = \frac{ln2}{t_{1/2}}$

  where $N_t =$ number of particles at time t, $N_0 =$ number of particles present at $t = 0$, $\lambda =$ decay constant, $t_{1/2} =$ time for half the radioactive amount to decay (ACSPH029)
- model and explain the process of nuclear fission, including the concepts of controlled and uncontrolled chain reactions, and account for the release of energy in the process (ACSPH033, ACSPH034)
- analyse relationships that represent conservation of mass-energy in spontaneous and artificial nuclear transmutations, including alpha decay, beta decay, nuclear fission and nuclear fusion (ACSPH032)
- account for the release of energy in the process of nuclear fusion (ACSPH035, ACSPH036)
- predict quantitatively the energy released in nuclear decays or transmutations, including nuclear fission and nuclear fusion, by applying: (ACSPH031, ACSPH035, ACSPH036)
  - the law of conservation of energy
  - mass defect
  - binding energy
  - Einstein's mass–energy equivalence relationship $E = mc^2$

Inquiry question: How is it known that human understanding of matter is still incomplete?

- analyse the evidence that suggests:
  - that protons and neutrons are not fundamental particles
  - the existence of subatomic particles other than protons, neutrons and electrons
- investigate the Standard Model of matter, including:
  - quarks, and the quark composition hadrons
  - leptons
  - fundamental forces (ACSPH141, ACSPH142)
- investigate the operation and role of particle accelerators in obtaining evidence that tests and/or validates aspects of theories, including the Standard Model of matter (ACSPH120, ACSPH121, ACSPH122, ACSPH146)

DISCHARGE TUBES AND CATHODE RAYS

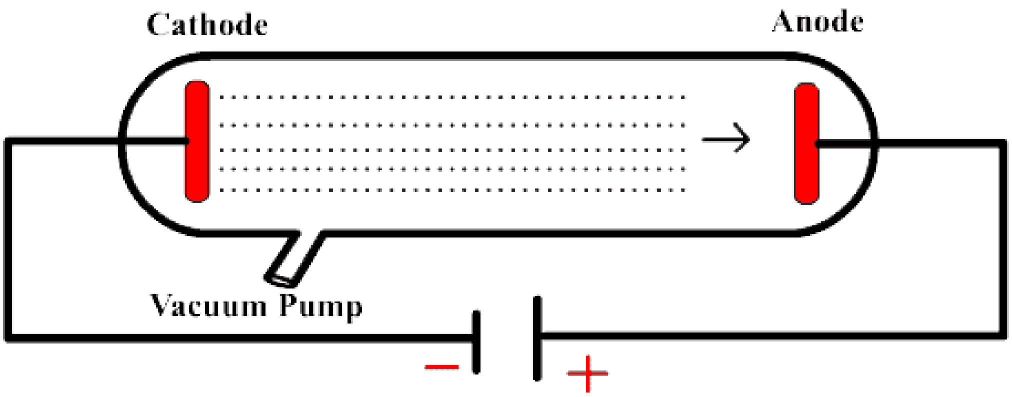
OUTCOMES

- investigate, assess and model the experimental evidence supporting the existence and properties of the electron, including:
  - early experiments examining the nature of cathode rays
  - Thomson's charge-to-mass experiment
  - Millikan's oil drop experiment (ACSPH026)

DISCHARGE TUBE

By the early 1700s, physicists were experimenting with **electricity** inside sealed glass tubes with **low air pressure**. They placed metal electrodes inside the discharge tubes and connected them to a voltage.

- The **positive electrode** was called the **anode**
- The **negative electrode** was called the **cathode**



Johann Hittorf in 1869 found that when **high voltage was applied across the very low air pressure**, rays of light appeared to travel from the cathode to anode. The ray was named **cathode rays** because they radiated from the cathode.

PRODUCTION OF CATHODE RAYS

Cathode rays are **streams of electron** flowing from the cathode towards the anode in a discharge tube. The discharge tubes are **cold vacuum tubes**.

The production of cathode rays in **cold discharge tubes** is outlined below:

- A small trace of ions and free electrons are present in the residual gas. There must be some pressure (gas).
- A **high voltage** is applied between the cathode and the anode, creating a **strong electric field** in the tube.
- The strong electric field **accelerates the ions** towards the cathode **and electrons** towards the anode
- As the ions and electrons accelerate through the tube, they **collide** with other molecules and ionise them, **releasing more free electrons**.
- As the electrons accelerate through the discharge tube, they **collide with the air molecules** and as a result **light is released**.

Low air pressure is required for cathode rays to be produced.

- As air is an insulator, it would prevent the flow of electrons between cathode to anode.
- To allow cathode rays to be produced, the air must be pumped out of the tube. Low pressure air will allow cathode rays to be produced
- The electrons can travel through low pressure air and collide with any air molecules along the way

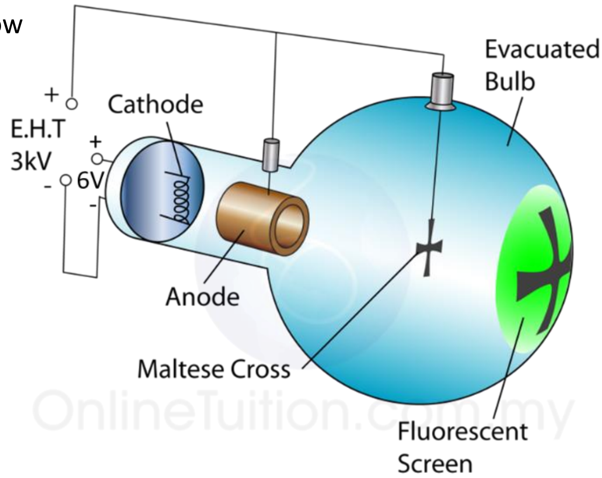
When electrons collide with atoms or molecules:

1) Low energy collision: Electron is deflected due to the gas being too close to each other. The accelerated electron does not have enough space between collisions to reach a kinetic energy that would emit visible light.
2) High energy collision: Some of the kinetic energy the electron acquired as a result to the high potential difference is converted into **light energy**, which makes the cathode ray visible.
3) Very high energy collisions: the electron **ionises** the atom or molecule, resulting in an ion and extra free electron

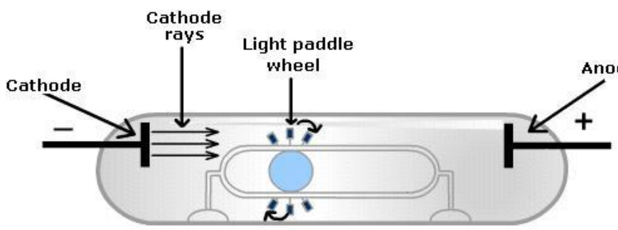
THE NATURE OF CATHODE RAYS

MALTESE CROSS BY JULIUS PLUCKER

When the tube was turned on, it cast a sharp cross-shaped shadow on the fluorescence on the back face of the tube. It could be concluded that cathode rays travel from the cathode to the anode in **straight lines**.



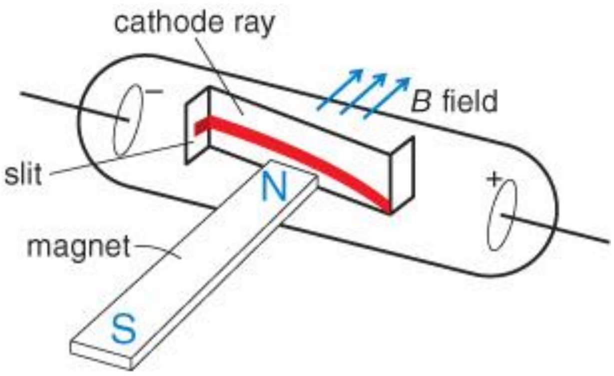
PADDLE WHEEL BY WILLIAM CROOKES



- Crooks concluded that the time that this showed that cathode rays **possess momentum** and therefore **mass** since they cause a wheel placed in their path to turn.
- This suggested that **cathode rays were made out of particles**.
- However, it was later concluded that the paddle wheel turned due to the **radiometric effect**

DEFLECTION BY MAGNETIC FIELDS BY WILLIAM CROOKES

It could be observed that cathode rays consists of negatively charged particles and therefore are deflected in a magnetic field according to the LHPR ($F_{magnetic} = qvB$).



- Cathode rays are **charged** and **undergo deflection** perpendicular to an applied magnetic field.
- Cathode rays are **negatively charged**. The direction of the force acting on a negatively charged particle **is opposite** to the direction of the force acting on a positively charged particle.

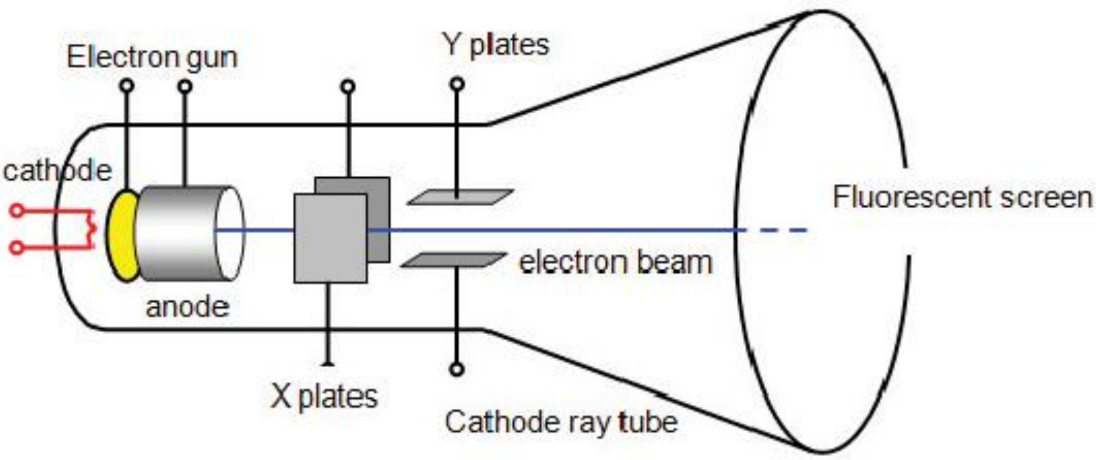
SUMMARY

Year	Experiment	Scientist	Conclusion
1869	Maltese Cross	Julius Plücker	Travels in straight lines. Wave?
1879	Paddle wheel	William Crookes	Have momentum and hence mass
1879	Deflection by magnetic fields	William Crookes	Must be charged
1895	Cathode rays carry negative charge	Jean-Baptiste Perrin	Negatively charged

THOMSON'S CHARGE-TO-MASS EXPERIMENT

J. J. Thomson showed that **cathode rays consisted of negatively charged** particles by **calculating the charge to mass ratio of cathode rays**. He is credited with:

- Discovery and identification of the electron
- Discovery of the first subatomic particle, showing that atoms are not indivisible



His experiment involved 3 main steps:

1) Acceleration of electrons
2) Deflection of electrons by magnetic fields
3) Deflection of electrons by both magnetic and electric fields

DEFLECTION OF ELECTRONS BY MAGNETIC FIELDS

The cathode ray was deflected with a magnetic field and the position of deflected ray with only the magnetic field acting was recorded. This allowed the radius of the curvature of the path of deflection due to the magnetic field to be determined.

$$F_{centripetal} = F_{magnetic}$$

$$\frac{mv^2}{r} = qvB$$

$$\frac{q}{m} = \frac{v}{Br}$$

DEFLECTION OF ELECTRON BY BOTH MAGNETIC AND ELECTRIC FIELDS

By varying magnetic field and electric fields until their opposing forces cancel each other out, leaving the cathode rays undeflected. By equating the magnetic field and electric force equations, he was able to determine the velocity of the cathode-ray particles (electrons).

$$F_{electric} = F_{magnetic}$$

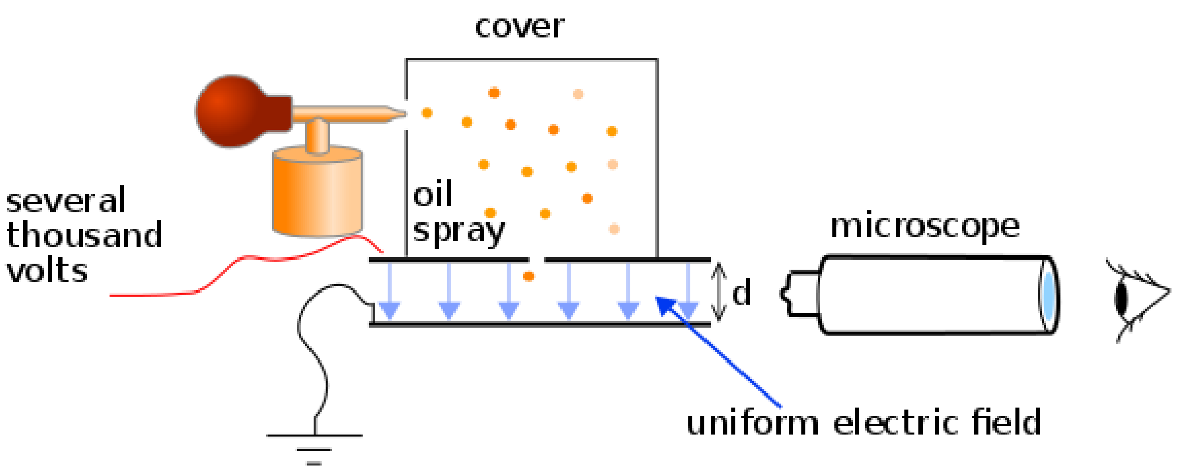
$$qe = qvB$$

$$v = \frac{E}{B}$$

$$\therefore \frac{q}{m} = \frac{E}{B^2 r}$$

MILLIKAN'S OIL DROP EXPERIMENT

Robert A. Millikan and Harvey Fletcher attempted to **measure the charge of an electron**, by measuring the charge of small, ionised droplets of oil.



1) Drops of oil are sprayed into a region above a pair of parallel plates
2) The drops become ionised either through friction with the nozzle of the spray, or by illuminating them with X-rays.
3) Drops can fall into the region between the plates. This region contains an **electric field** set up by a **voltage across plates**.
4) Once between the plates the drops can be viewed through a microscope. The drops can be accelerated up or down by adjusting the voltage on the places, and hence the electric field.
5) The **voltage** was adjusted such that some of the drops were **suspended**.

$$F_{electric} = F_{gravitational}$$

$$qE = mg$$

$$q = \frac{mg}{E}$$

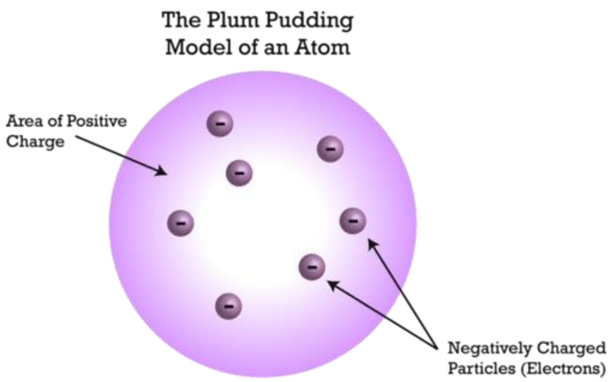
Millikan and Fletcher determined the **terminal velocity** of the oil droplets and made careful measurements of their diameters.

- This allowed them to determine the **mass of their oil drops**.
- Knowing the mass, the charge could be found as well

THOMSON'S MODEL OF THE ATOM

- Thomson had shown that particles smaller than atoms existed
- He concluded that electrons must exist inside atoms and that atoms must therefore be divisible into smaller parts
- Thomson then had to incorporate electrons into the structure of the atom.

In 1911, Thomson proposed the **Plum Pudding Model of the Atom**.



He suggested that an atom is **sphere of positively charged** in which electrons are embedded like raisins in a plum pudding.

OUTCOMES

- investigate, assess and model the experimental evidence supporting the nuclear model of the atom, including:
  - the Geiger−Marsden experiment
  - Rutherford's atomic model
  - Chadwick's discovery of the neutron (ACSPH026)

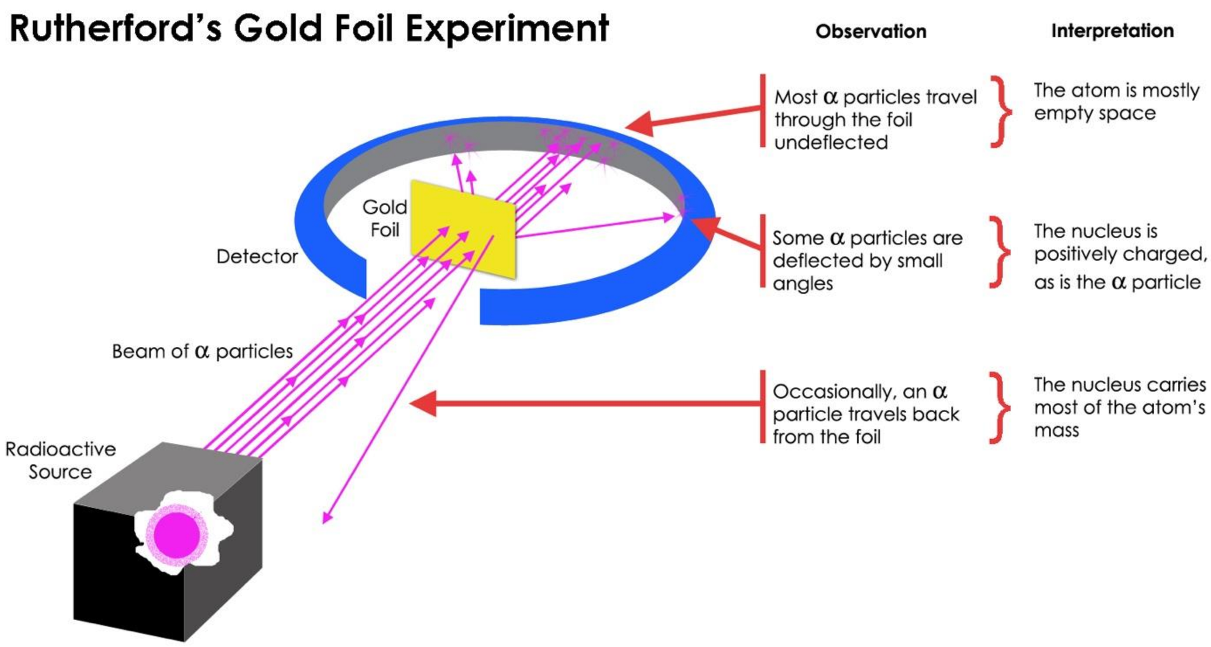
RUTHERFORD'S ALPHA PARTICLES

Rutherford's work was the identification of **alpha particles** which where given off by some radioactive elements.

An alpha particle is a **helium ion** ${}^4_2He$. Alpha particles are emitted from radioactive elements with **very high speeds** (0.1c). Rutherford recognised the value of alpha particles in **probing the structure of atoms**.

GEIGER-MARSDEN "GOLD FOIL EXPERIMENT"

The aim was to **investigate the plum pudding model** by using alpha particle scattering.



- Alpha particles were fired through a piece of gold foil and used a ZnS detector to detect the scatter of alpha particles and their location
- Gold was used as it is malleable and could be beaten into a film comprising of 400 atoms.

THEORETICAL PREDICTION

Rutherford expected that the gold atoms would not affect the positive alpha particles for the following reasons:

- The positive charge of the gold atoms was **uniformly distributed over a large volume**
- The negative electrons were **too light** to deflect the alpha particles.

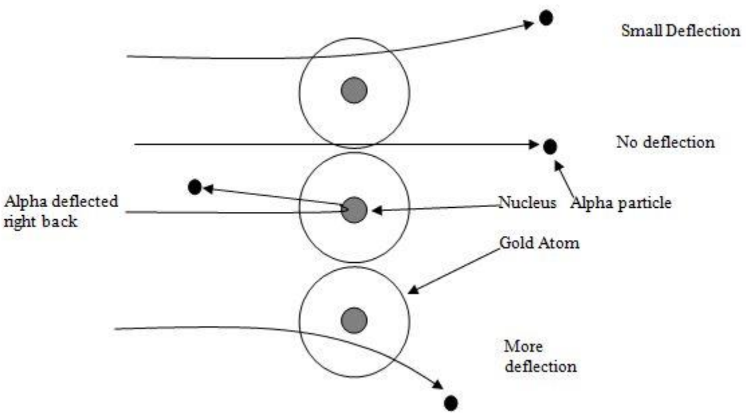
Hence there would be **no concentration of charge or mass** large enough to deflect the alpha particles

EXPERIMENTAL RESULTS

- 1 in 1800 alpha particles were deflected by a minor (10 degrees) from their path
- 1 in 8000 alpha particles deflected greater than 90 degrees

CONCLUSION

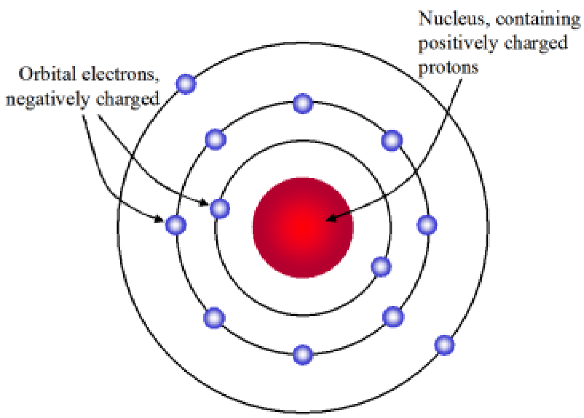
Rutherford concluded that there must be a **small, massive, positively charged core** in the atoms that causes the alpha particles to deflect.



RUTHERFORD'S PLANETARY MODEL OF THE ATOM

Rutherford proposed that each atom has a **dense central core**, which he called the **nucleus**.

- The nucleus is a central region that is **very small** relative to the total size of the atom
- The nucleus contains virtually all the mass (99.9%) and all the **positive charge** of the atom
- The positive charge of the nucleus is to be balanced by the negative charge of the electrons outside the nucleus to maintain electrical neutrality.
- The **electrons orbit around the nucleus in uniform circular motion**.



Rutherford's new model assumed that electrons **orbited** the nucleus. This idea presented some problems that collided with already established Physics:

1) The electrons in orbits are **constantly accelerating** (circular motion) and should be giving off EMR
2) The electrons should be losing energy and spiral into the nucleus causing the atoms to collapse
3) It could not explain the emission line spectrum for hydrogen or any other element

PROTONS AND NEUTRONS

Rutherford conducted an experiment that proved **hydrogen nuclei were present in heavier nuclei**.

- Rutherford observed that **hydrogen nuclei were produced when alpha particles were fired at nitrogen**.
- From this, he concluded that the nitrogen nuclei (and hence all nuclei) contained hydrogen nuclei within them

$$^{14}_{7}N + ^{4}_{2}He \rightarrow ^{17}_{8}O + ^{1}_{1}H$$

Rutherford proposed that the **nucleus of hydrogen** consisted of a **fundamental building block** which made up all other nuclei.

- He named this particle the **proton** in 1920, meaning 'first' in Greek.

PROBLEMS WITH CALCULATING THE MASS OF NUCLEI

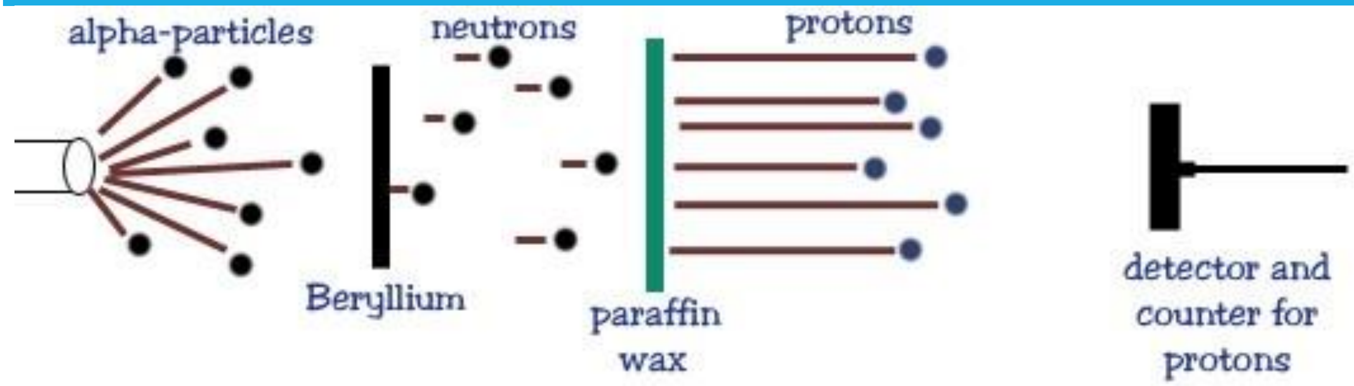
In alpha particle scattering experiments, there was a problem when the experimental data were used to calculate the masses of the nuclei.

The mass and charge of the proton were already known, and it was believed that heavier nuclei contained more protons. However, the results of the scattering experiment showed that:

- The **charge** of a nuclei was **equal** to the charge of the protons contained, as expected
- However, the **mass** of the nucleus was **roughly double** the mass of the protons contained

In 1920, Rutherford proposed that the nucleus might contain a second **uncharged** particle with a mass close to that of a proton.

JAMES CHADWICK'S DISCOVERY OF NEUTRONS



- Polonium was used as an alpha source, and the emitted alpha particles were directed at a piece of beryllium
- When struck, the beryllium emitted the mysterious "neutral rays" which where highly penetrating
- Paraffin wax (really long hydrocarbon chain) was placed in the path of the mysterious radiation. When the rays hit the paraffin wax, they knocked hydrogens (protons) out of it.
- Due to their charge, the protons were able to be defected by a **Geiger counter** and their energy and speed could be calculated

Chadwick conducted similar experiments where the radiation was fired at **different atoms with different masses**

- He measured the speed of the atoms that were ejected

- He analysed the **collisions** between the radiation and these different atoms, considering the **conservation of energy and momentum**
- In 1932, Chadwick was able to prove the existence of a **neutral particle** with a **mass close to that of a hydrogen atom.**

EXPLANATION OF CHADWICK'S EXPERIMENT

The following reaction was taking place between alpha particles and the beryllium:

$$^{9}_{4}Be + ^{4}_{2}He \rightarrow ^{12}_{6}C + ^{1}_{0}n$$

The neutrons travelled at high speeds and collided with the nuclei in the target

- The nuclei could be easily detected as they were charged (protons). Their mass was known and their speed could be measured
- This allowed their **energy** and **momentum** to be determined.
  - This was necessary as neutrons have **no electric charge** and therefore don't readily ionise atoms. This makes it extremely difficult to detect, but if they were made to collide with small nuclei, the energy and momentum can be measured and used to calculate the mass of the neutron.

BOHR'S ATOMIC MODEL OF THE ATOM

OUTCOMES

- assess the limitations of the Rutherford and Bohr atomic models
- investigate the line emission spectra to examine the Balmer series in hydrogen (ACSPH138)
- relate qualitatively and quantitatively the quantised energy levels of the hydrogen atom and the law of conservation of energy to the line emission spectrum of hydrogen using:
  - $E = hf$
  - $E = \frac{hc}{\lambda}$
  - $\frac{1}{\lambda} = R\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right]$ (ACSPH136)

PROBLEMS WITH RUTHERFORD'S ATOMIC MODEL

There are **two** basic difficulties with Rutherford's planetary model of the atom.

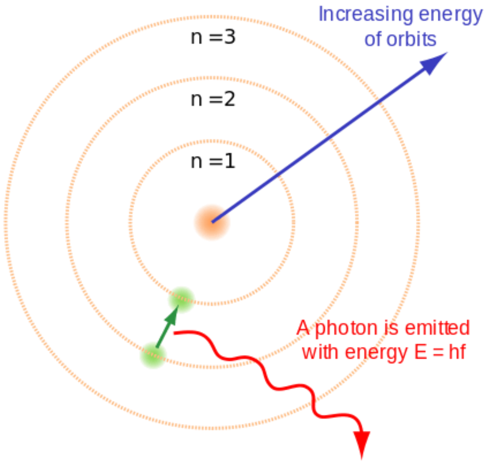
- It cannot explain why **atoms are stable**
  - According to Maxwell's theory of electromagnetism, the **accelerating electrons** should emit electromagnetic waves. This means that the electrons should be losing energy. The atom should collapse as the **electron plunges** into the nucleus.
- It cannot explain the **emission and absorption spectra** of atoms.
  - Hydrogen absorb only specific wavelengths, and emit light only at the wavelengths they absorb. These are called **line spectra**.
    - Rutherford's model had **no structure for the electrons**, they were free to orbit at any radius, hence they should be able to absorb and emit radiation at **any wavelength**

The **problems** with Rutherford's model arise due to its **classical nature**. It treats the motion of electrons using Newton's Laws and makes predictions based on Maxwell's theories, both of which are classical physics principles.

THE BOHR ATOMIC MODEL

BOHR'S POSTULATES

Postulates	Explanation
Only certain electron orbits are stable	These stable orbits are ones in which the electron does **not radiate**. Hence the energy is fixed, and **classical physics does not hold**.
Radiation is absorbed or emitted when electrons transition from one stable orbit to another	This transition cannot be visualised or treated classically. The electrons cannot exist between two stable orbits – it will exist in one orbit and then instantaneously transition to another **The difference in energy between the two stable orbits must be absorbed or emitted as a single photon**, following Planck's and Einstein's theories: $$E_i - E_f = hf = \frac{hc}{\lambda}$$ Where E_i is the energy of the initial state, E_f is the energy f the final state. ($hf < 0$ indicates absorption, $hf > 0$ indicates emission)
The stable electron orbits have quantised angular momentum.	The size of the allowed electron orbits is determined by a **quantisation condition** imposed on the **electron's orbital angular momentum.** The allowed orbits are those for which the electron's orbital angular momentum about the nucleus is an integral number of $\frac{h}{2\pi}$ $$L_n = mvr = \frac{nh}{2\pi}$$ For $n = 1, 2, 3, ...$



- Each **stable orbit** is represented by a circle
- Each **stable orbit** is uniquely labelled with the **integer** n that determines the orbital angular momentum $\left(L = mvr = \frac{nh}{2\pi}\right)$
- The integer n is called the **quantum number**, since it counts how many quanta (packets of $\frac{h}{2\pi}$) of angular momentum each orbit has.

Bohr calculated the **energy** an electron must have in each stable orbit of a hydrogen atom.

- The energy of each stable orbit according to Bohr's model is given by:

$$E_n = -\frac{13.6eV}{n^2}$$

- The energy is negative due to the **large negative potential energy**
- The lowest energy level is $n = 1$, and is knowna s the **ground state**. It is the closest to the nucleus and hence has the lowest potential energy
- Higher levels of n are known as **excited states**
- To remove an electron (ionise) it takes 13.6 eV

EMISSION SPECTRUM

The emission spectrum for any gaseous element at its ground state can be produced by:

- Heating substances until they glow
- Applying a high voltage to gases in glass tubes causing the gas to glow
- Burning salts in gas flames causing the salt to produce a bright flash of light

BOHR'S EXPLANATION

- The electron moves in a circular orbit around the nucleus
- The force keeping the electron in uniform circular motion is the **electrostatic force of attraction**
- A number of allowable orbits of different radii (stationary states) where electrons can orbit
- The electrons will occupy the lowest energy orbit available
- An electron can jump to higher energy levels by absorbing photons (excitation). The energy absorbed is equal to the difference between the final and initial energy levels ($\Delta E = E_f - E_i = hf$)
- EMR is emitted when an excited electron relaxes and falls from a higher energy level to a lower energy level.

$$E = hf = \frac{hc}{\lambda}$$

Where:

- E: Energy of the photon (j)
- h: Planck's constant
- f: frequency
- λ: wavelength
- c: speed of light

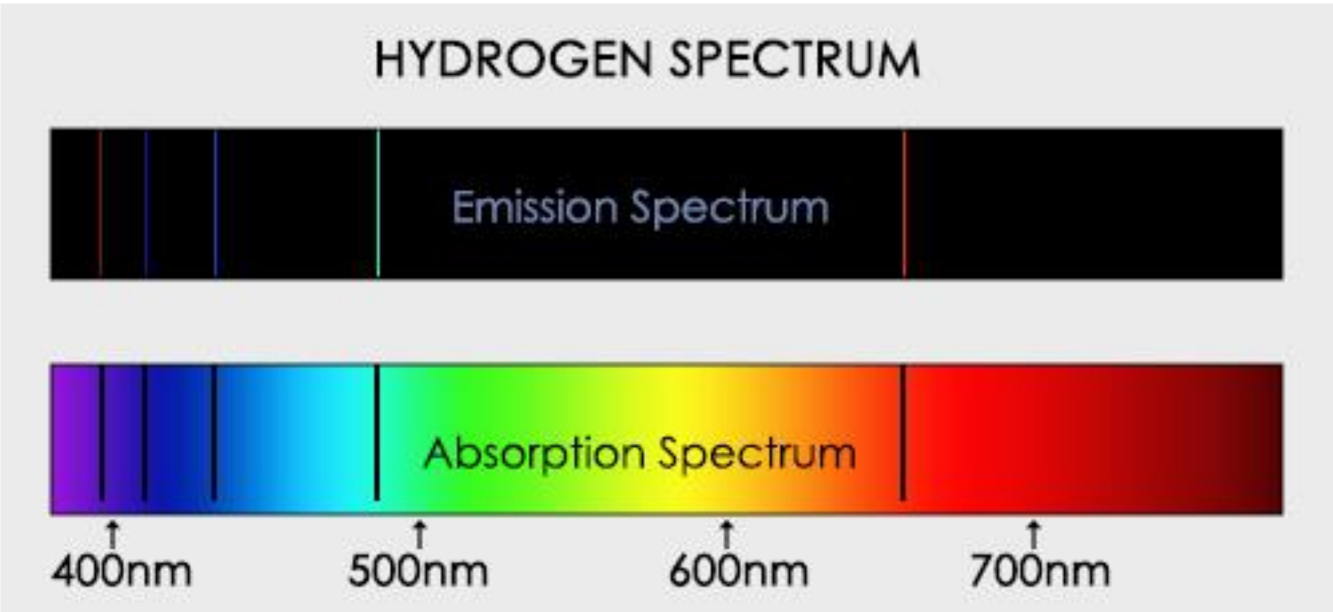
BOHR AND THE EMISSION SPECTRUM OF HYDROGEN

The hydrogen atoms gain energy from heating or an electrical current. This excites electrons to jump into a higher energy level. It will then relax and drop from the higher energy level to one of the lower levels. During this process, it will emit a photon with energy that is equal to the difference in energy between the two levels. The given off photons can be plotted on an emission spectrum.

Bohr noticed that:

1) The absorption spectrum of hydrogen showed it was capable of absorbing a small number of frequencies of light. Therefore, the energies of light absorbed must be **quantised**.
2) The emission spectrum of hydrogen showed that hydrogen atoms could only emit quanta (energy) with the exact same energy values it was able to absorb.
3) Hydrogen have an ionisation energy of 13.6 eV. This is the energy needed to remove an electron from the hydrogen.
4) Photons of light with all energies above the ionisation value are continually absorbed.

If the energy of a photon is greater than the energy of ionisation, the excess energy becomes excess **kinetic energy** for the released electron. Only **incident photons** carrying just the right amount of energy can raise an electron to an allowed level.



EXAMPLE QUESTION 1

Describe Niels Bohr's model of the atom and outline the **impacts** it had on the development of the atom.

Rutherford's model of the atom pictured electrons in circular orbits at different distances from the nucleus. But this model could not account for the stability of the orbits. (A centripetally accelerating electron should, according to Maxwell's equations, continually radiate EMR, lose energy and spiral into the nucleus) or the discrete nature of atomic emission and absorption spectrum.

Bohr's model addressed these short comings. Bohr proposed that an electron can orbit a nucleus only at certain allowed angular momentum and energies (His 3rd postulate: $L_n = mvr = \frac{nh}{2\pi}$ $(n = 1, 2, 3, \dots)$ $E = -\frac{13.6eV}{n^2}$ for hydrogen) and that these orbits are somehow exempt from the requirement of Maxwell's equations that accelerating charges continually emit EMR.

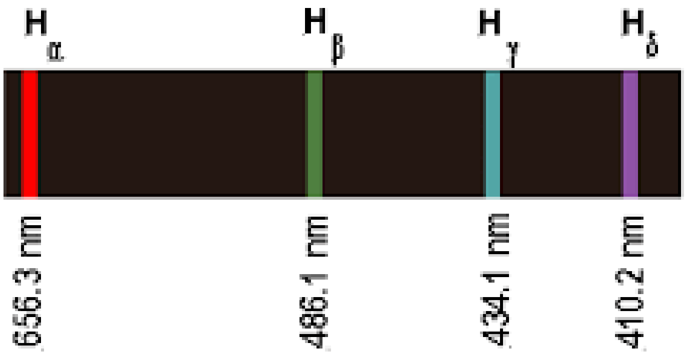
When an electron relaxes from a higher to lower energy level, it emits a single photon (His 2nd postulate)

$$E_{photon} = hf = -\Delta E_{electron}$$

Because only certain transitions ($\Delta E_{electron}$) are possible, only certain frequencies of light are emitted (or absorbed, when the electron is excited to higher allowed energy): $f = \frac{|\Delta E_{electron}|}{h}$

THE BALMER SERIES

Johann Jacob Balmer found an empirical formula that correctly predicted the wavelengths of **four visible emission lines** of hydrogen.



Spectrogram of visible lines in the Balmer series of hydrogen as obtained with a constant-deviation spectrograph.

(Typically, the other way around)

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

Where:

- R_H is the Rydberg constant, $(1.097 \times 10^7\ m^{-1})$
- $n = 3,\ 4,\ 5, 6, \dots$

The Blamer series for the hydrogen atom corresponds to **electronic transitions that terminate in the state of quantum number n=2.**

Colour of the emission line	Name of the emission line	Electron transition	Wavelength
Red	H_α	$n = 3\ to\ n = 2$	$656.3\ nm$
Green	H_β	$n = 4\ to\ n = 2$	$486.1\ nm$
Blue	H_γ	$n = 5\ to\ n = 2$	$434.1\ nm$
Violet	H_δ	$n = 6\ to\ n = 2$	$410.2\ nm$

RYDBERG'S EQUATION

Johannes Rydberg showed that all the emission series could be predicted using a generalised version of Balmer's equation:

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

Where:

- R: Rydberg constant $(1.097\ \times 10^7\ m^{-1})$
- n_i and n_f are integers where $n_i > n_f$

BOHR'S DERIVATION

From Bohr's first postulate:

$$L_n = mvr = \frac{nh}{2\pi}$$

The energy of each stable orbit is quantised and given by:

$$E_n = -\frac{13.6eV}{n^2}\ for\ n = 1, 2, 3, \ldots$$

$$\therefore E_i = \frac{1}{n_i^2} \times 13.6eV\ \&\ E_f = \frac{1}{n_f^2} \times 13.6eV$$

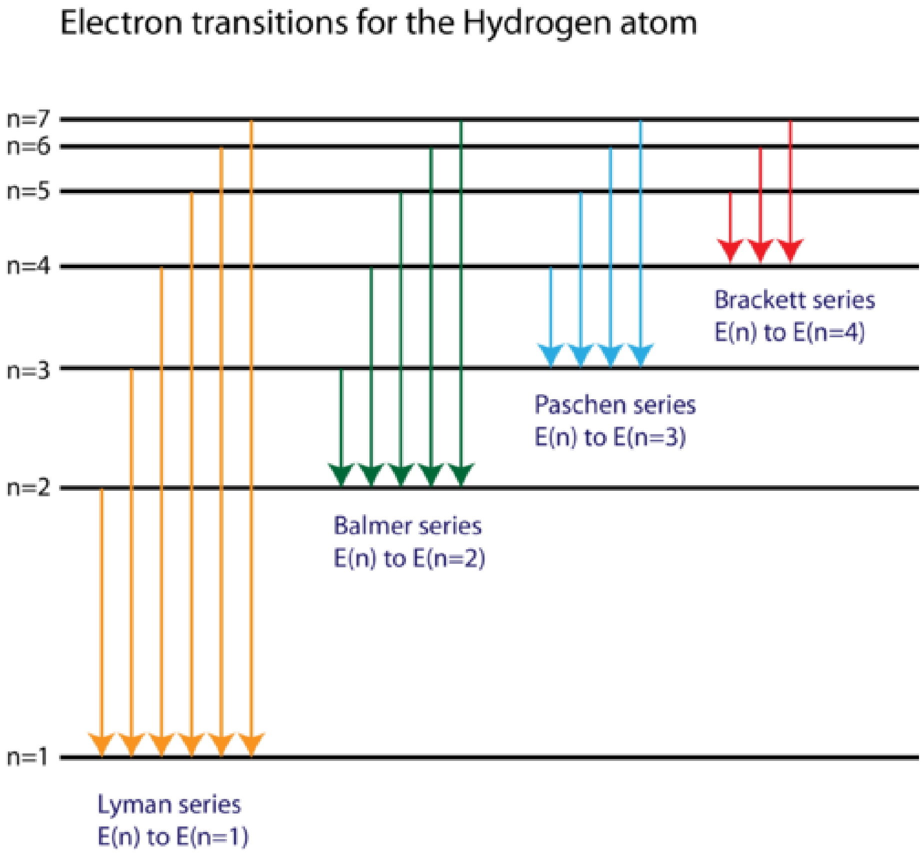
From Bohr's second postulate:

$$E = E_i - E_f = -13.6eV\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

From Einstein's photon theory:

$$E = \frac{hc}{\lambda} => \frac{1}{\lambda} = \frac{E}{hc}$$

$$\frac{1}{\lambda} = \frac{13.6eV}{hc}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) => \frac{1}{\lambda} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$



LIMITATIONS OF BOHR'S MODEL OF THE ATOM

The underlying reason why Bohr's model has limitations is that it is **oversimplified**.

- It uses one quantum number n, to describe the stable orbits. The correct quantum model requires 4 quantum numbers.
- Since it was a simple model, it could only explain the simplest atom: hydrogen.

THE SPECTRA OF LARGER ATOMS

Bohr's analysis worked well for **single electron** atoms or ions. It could not explain the spectra of larger atoms which are more complex. A complex atom has more than one electron and these electrons will interact. Orbital electrons experiences forces from other electrons (due to their closeness in the atom). Bohr's model does not account for the additional forces.

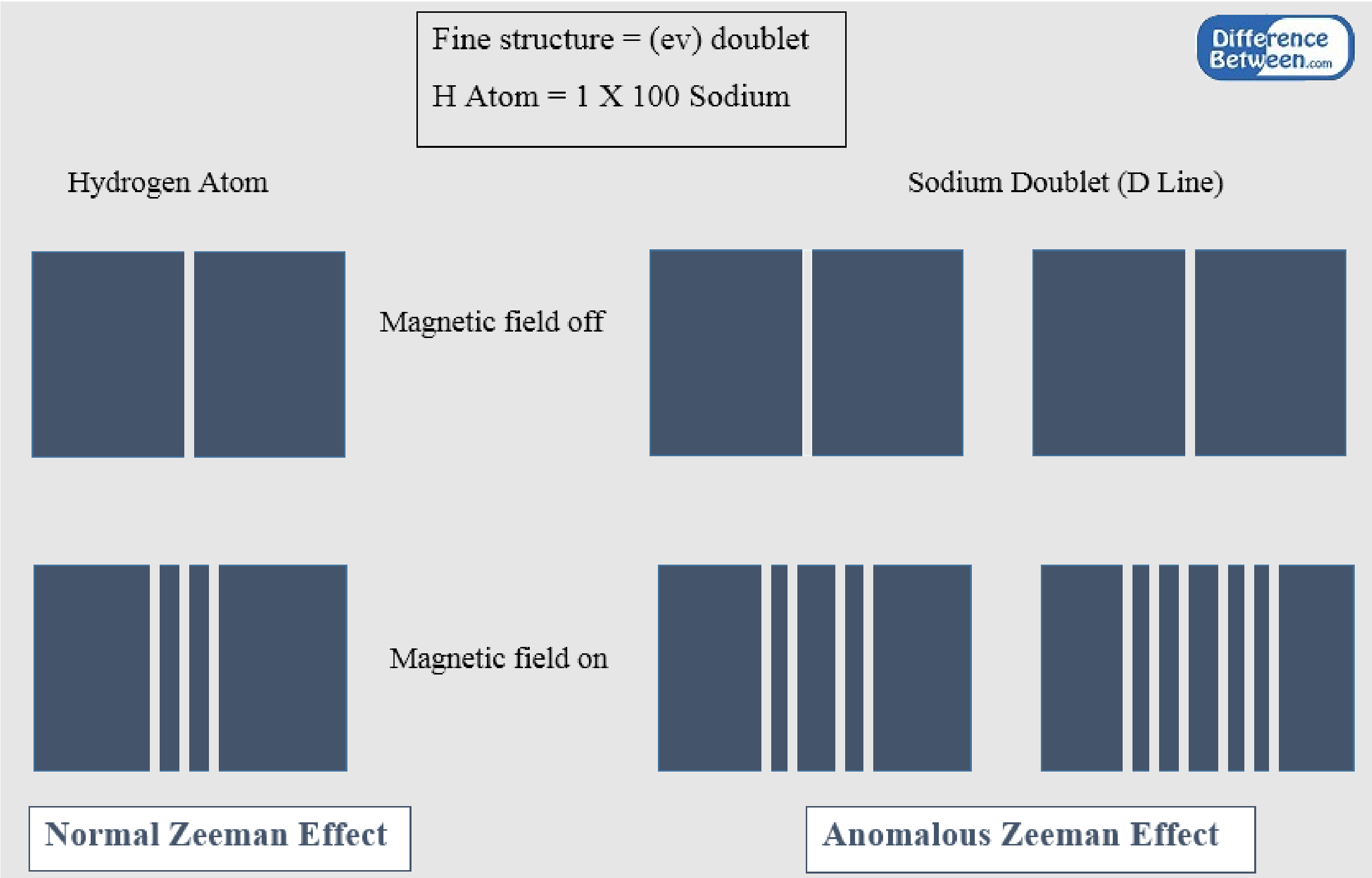
His quantisation of angular momentum worked for a single electron. Bohr did not know how to adjust his model for several electrons.

ZEEMAN EFFECT

When spectral lines were recorded with a discharge tube placed in a **magnetic field**, the recorded spectral lines were split into **several finely separated lines**.

This implies that energy levels are split. This is not consistent with stable energy levels in Bohr's model.

The magnetic field can create new allowed energy levels for an electron in an atom. Because Bohr didn't initially understand why his quantisation of angular momentum was accurate, he did not know how to adjust the model to account for an external magnetic field.



QUANTUM MODEL OF THE ATOM

OUTCOMES

- investigate de Broglie's matter waves, and the experimental evidence that developed the following formula:
  – $\lambda = \frac{h}{mv}$ (ACSPH140)
- analyse the contribution of Schrödinger to the current model of the atom

DE BROGLIE'S MATTER WAVE

Louis Victor de Broglie in 1923 postulated that because **photons have wave and particle characteristics**, **all forms of matter have wave as well as particle properties**. According to his hypothesis:

- Electrons have dual particle/wave nature
- In analogy with photons having a wavelength, he proposed that the wavelength of a particle is related to momentum by:

$$\lambda_{particle} = \frac{h}{p} = \frac{h}{mv}$$

Where:

- m: mass of particle (kg)
- v: velocity of particle (m/s)
- h: Planck's constant
- λ: wavelength (m)

EXPERIMENTAL EVIDENCE OF MATTER WAVES

De Broglie's proposal that any kind of particle exhibits both wave and particle properties was first regarded as pure speculation.

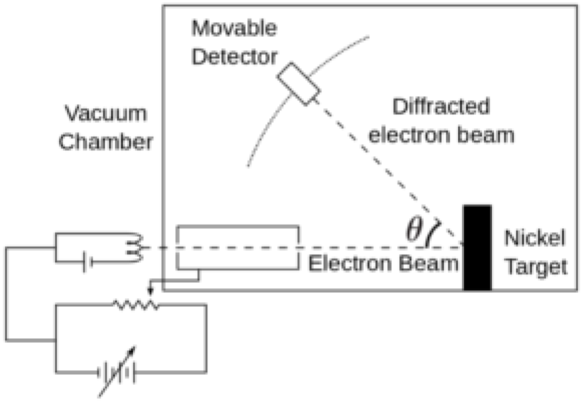
- **Diffraction** is a wave property whereby waves are able to bend around an obstacle and spread out. This often results in the superposition of waves.
- **Interference** is a wave characteristic and results from the superposition of two or more waves, giving rise to a new wave pattern. There are two types of interference, **constructive** and **destructive**.

If particles such as electrons had **wave-like properties**, then under the correct conditions they **should exhibit interference phenomena**.

DAVISSON-GERMER EXPERIMENT

The experiment confirmed De Broglie's hypothesis that a particle of momentum $p = mv$ will like a wave with wavelength $\lambda = \frac{h}{mv}$.

Davisson and Germer were investigating the **scattering of low energy electrons** from nickel target in a vacuum.



- Electrons from heated filament are accelerated by an adjustable potential difference. The resultant beam, made up of electrons whose kinetic energy is 54 eV, travels to a nickel crystal.
- The beam 'reflected' from the crystal surface enters the detector and is recorded as a current.

Experiments showed that the **scattered electrons exhibited intensity of maxima and minima at specific angles**. The electron was being scattered by different layers within the crystal lattice of the target and were undergoing interference. When they calculated the diffraction pattern of 0.14nm which was consistent to De Borglie's hypothesis

ELECTRON WAVES IN THE ATOM

DE BROGLIE'S EXPLANATION OF THE STABLE ELECTRON ORBITS

Bohr's model had many **shortcomings** and **problems** such as:

- No justification for why the stable orbits were stable (not continually radiating light and spiralling into the nucleus)
- No justification for the quantisation of angular momentum

De Broglie used **wave theories** of matter to **resolve problems with the stability of electron orbits in the Bohr atom**. De Broglie used **interference** to resolve the problem, postulating that **stable orbits** are **standing waves**.

- An electron can orbit an atomic nucleus only if the **circumference of the orbit is a whole number of electron wavelengths**, so that it forms a **standing wave**. This way the electron is not accelerating.
- An orbit of radius r will be required to satisfy the condition: $n\lambda = 2\pi r$

If the electron wave travels around the circular orbit, it will return to the same place and superimpose with itself, leading to interference.

- If it ravels a **whole number of wavelengths** around the circumference, it will be **in phase** with itself at every position, hence it will **constructively interfere**. This is what **forms the standing wave**.
- If it does **not travel a whole number of wavelengths** around the circumference it will eventually be **out of phase** with itself. This would lead to destructive interference and **will not form a standing wave**.

All of the allowed states (stable orbits) of the electron are standing waves states on a circumference, each with its own wavelength, speed and energy.

DE BROGLIE'S HYPOTHESIS AND QUANTISATION OF ANGULAR MOMENTUM

The condition for the stability of an electron orbit is given by: $n\lambda = 2\pi r$

Substituting de Broglie wavelength expression: $n\left(\frac{h}{mv}\right) = 2\pi r \Rightarrow mvr = \frac{nh}{2\pi}$

TOMSON'S DIFFRACTION EXPERIMENT

Thomson produced a diffraction pattern by passing a beam of electrons through a tiny crystal. He then repeated the experiment using X-rays of the **wavelength** in place of the electron. The diffraction pattern was almost identical to the electron's pattern. He therefore concluded that electrons and X-rays have a similar wavelength (but not speed).

SCHRODINGER'S EQUIATION

In 1926, Erwin Schrodinger published his equation to **describe wave mechanics**. The Schrodinger equation for a hydrogen atom can be solved to find the allowed states for the electron.

- These solutions are **discrete** and correspond to **standing waves**
- However, these are now **three-dimensional standing waves** which reflect the fact that electrons are free to move in three dimensions.

HEISENBERG'S UNCERTAINTY PRINCIPLE

The **uncertainty principle** relates to the ability to make measures with a certain precision.

In classical mechanics:

- There is no fundamental barrier to an ultimate refinements of apparatus and/or experimental procedures.
- It would be possible, in principle, to make any measurement with arbitrarily small uncertainty.

Quantum theory predicts that:

- It is **fundamentally impossible** to make **simultaneous** measurements of a particle's position and velocity with **infinite accuracy**
- If you were to measure the position and velocity of a particle at any time, you will always be faced with uncertainties in your measurements.

$$\Delta x \, \Delta p \geq \frac{h}{4\pi}$$

$$\Delta E \, \Delta t \geq \frac{h}{4\pi}$$

- Δx: uncertainty in position
- Δp: uncertainty in momentum
- ΔE: uncertainty in energy

QUANTUM MODEL OF THE ATOM

- Electrons have a wave nature as proposed by De Broglie. Their wave nature is described by Schrodinger's equation.
- Electrons in an atom occupy discrete quantum states that are described using 4 quantum numbers.
- Electrons form three-dimensional standing waves around the nucleus. These are the solutions to Schrodinger's equation and are described by three quantum numbers
- Electrons have spin, which is quantised as up or down, described by the 4^{th} quantum number
- Each quantum state holds at most one electron due to the Exclusion Principle. Hence one electron fills each quantum state, starting from the lowest energy
- The electron can be found anywhere within the orbital. We cannot know the exact position of the electron due to the uncertainty principle.

NUCLEAR PHYSICS

OUTCOMES

- analyse the spontaneous decay of unstable nuclei, and the properties of the alpha, beta and gamma radiation emitted (ACSPH028, ACSPH030)
- examine the model of half-life in radioactive decay and make quantitative predictions about the activity or amount of a radioactive sample using the following relationships:
  - $N_t = N_o e^{-\lambda t}$
  - $\lambda = \frac{ln2}{t_{1/2}}$

  where N_t = number of particles at time t, N_0 = number of particles present at $t = 0$, λ = decay constant, $t_{1/2} =$ time for half the radioactive amount to decay (ACSPH029)

PROPERTIES OF NUCLEI

A **nucleon** is the central part of an atom, the nucleus, that consists of protons and neutrons.

Isotopes have the same chemical properties but different physical properties such as density and volume. E.g:

- ^1_1H: Hydrogen
- ^2_1H: Deuterium
- ^3_1H: Tritium

CHARGE AND MASS

Particle	Charge (C)	Charge relative to Electron	Mass (kg)	Mass (u)
Proton	1.602×10^{-19}	1	1.673×10^{-27}	1.007
Neutron	0	0	1.675×10^{-27}	1.0084
Electron	-1.602×10^{-19}	-1	9.109×10^{-31}	0.000548

The **unified mass unit** (also called **atomic mass unit**) is used to expressed atomic masses instead of kilograms:

- The unified mass unit is defined such that the mass of a $^{12}_{6}C$ atom is exactly $12u$
- $1\,u = 1.661 \times 10^{-27} kg$

NUCLEAR REACTIONS

Nuclei can undergo nuclear reactions that will **redistribute the protons and neutrons** in the nuclei involved and will result in the **release or absorption of energy**.

$$^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

Two rules must be obeyed when writing nuclear reactions:

- **Conversation of charge**
- **Conversation of mass number**

RADIOACTIVE DECAY

NUCLEAR TRANSMUTATION

Nuclear transmutation is the process of **one element changing into another element** through nuclear reactions.

Natural transmutation is when radioactive isotopes spontaneously decay over a period of time and transform into other more stable isotopes. An example of the decay of uranium is:

$$^{238}_{92}\mathrm{U} \rightarrow \, ^{234}_{90}\mathrm{Th} + \, ^{4}_{2}\mathrm{He}$$

Artificial transmutation occurs when a nuclear reaction is triggered by one of the reactants. For example:

$$^{9}_{4}Be + \, ^{4}_{2}He \rightarrow \, ^{12}_{6}C + \, ^{1}_{0}n$$

RADIOACTIVITY

The **spontaneous** breakdown of an isotope into a new isotope by the emission of alpha, beta and/or gamma radiation is known as **natural radioactivity**. An isotope that is radioactive is called a **radioisotope**.

Type of Radiation	Alpha	Beta-plus & Beta-minus	Gamma
Particle Emitted	Helium Nuclei $^{4}_{2}He$	$^{0}_{+1}\beta$ or $^{0}_{-1}\beta$	$^{0}_{0}\gamma$ photon
Charge	+2	+1 or -1	No charge
Penetration	Stopped by 5cm of air or 0.5 mm of paper	Stopped by 0.5cm of aluminium	Intensity halved by 10cm of lead
Energy (Typical)	$10\ MeV$	$0.03 - 3\ MeV$	$1\ MeV$
Ionisation Effect	High	Medium	Low
Deflected by magnetic (B) field	Yes	Yes	No
Mass	Heavy	Light	None
Speed in Vacuum (c)	10%	90%	100%

ALPHA DECAY

Alpha decay involves the emission of an **alpha particle** ($^{4}_{2}He$).

$$^{A}_{Z}\mathrm{X} \rightarrow \, ^{A-4}_{Z-2}\mathrm{Y} + \, ^{4}_{2}\alpha$$

$$^{238}_{92}\mathrm{U} \rightarrow \, ^{234}_{90}\mathrm{Th} + \, ^{4}_{2}\alpha$$

Uranium-238 → Thorium-234 + Alpha Particle

ETA-MINUS DECAY

Occurs when an electron is emitted from the nucleus of a radioactive atom. It can be written as ${}_{-1}^{0}\beta$. This typically occurs when there are too many **neutrons** in a nucleus. A neutron spontaneously changes into a **proton** releasing a **beta-minus particle (electron)** and an uncharged massless antimatter particle called **antineutrino $\overline{v}$**.

$${}_{6}^{14}C \rightarrow {}_{7}^{14}N + {}_{-1}^{0}\beta + {}_{0}^{0}\overline{v}$$

$${}_{0}^{1}n \rightarrow {}_{1}^{1}p + {}_{-1}^{0}\beta + {}_{0}^{0}\overline{v}$$

BETA-POSITIVE DECAY

Similar to beta minus decay, it occurs when the nucleus has **too many protons**. The proton spontaneously changes into a neutron and emits a neutrino and beta-positive particle.

$${}_{1}^{1}p \rightarrow {}_{0}^{1}n + {}_{+1}^{0}\beta + {}_{0}^{0}v$$

GAMMA DECAY

After a radioisotope has emitted alpha or beta particles, the daughter nucleus usually has excess energy. The protons and neutrons rearrange slightly and offload this excess energy by releasing a gamma ray ${}_{0}^{0}\gamma$.

$${}_{\mathrm{Z}}^{\mathrm{A}}\mathrm{X}^{*} \rightarrow {}_{\mathrm{Z}}^{\mathrm{A}}\mathrm{Y} + {}_{0}^{0}\gamma$$

HALF-LIFE AND ACTIVITY

HALF-LIFE

Radioisotopes decay by emitting **ionising nuclear radiation**. Therefore, the quantity of a radioactive sample will **decrease** over time.

$$N_t = N_0 e^{-\lambda t}$$

Where:

- N_0: Initial undecayed nuclei sample
- λ: Decay constant (s^{-1})
- N_t: Number of remaining nuclei at time t

$$N_t = N_0 \left(\frac{1}{2}\right)^n$$

$$A = A_0 \left(\frac{1}{2}\right)^n$$

$$A = \lambda n$$

- n: Number of half-life elapsed
- A: activity

$$\lambda = \frac{\ln(2)}{t_{1/2}}$$

ACTIVITY

The activity is the **number of decays per second** that occurs in a sample of radioactive material.

- It is measured in **becquerel (Bq)** where 1 Bq = 1 decay per second

A **decay series** is when a radionuclide decays and its daughter nuclei is unstable and undergoes further decay and so on until it becomes a stable nucleus.

THE STRONG NUCLEAR FORCE

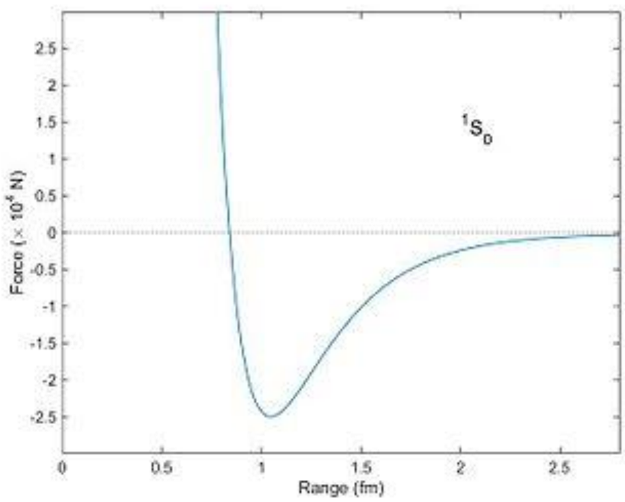
The nucleus contains **positive protons** and **neutral neutrons**, collectively referred to as **nucleons**. There must be a force of repulsion between the positively charged proton. Therefore, some additional force must be present to balance the proton's electrostatic repulsion.

STRONG NUCLEAR FORCE: THE NUCLEAR BINDING FORCE

There is an additional force **acting between nucleons** called the **strong nuclear force**.

- It is responsible for the stability of the nuclei
- This force represents the "glue" that holds the nuclear constituents (nucleons) together and overcomes the electrostatic repulsion.

For separations of about $10^{-15}\ m$ (1 femtometer), the strong nuclear force is orders of magnitude stronger than the electromagnetic force. **This holds the nucleons together in the nucleus**. The strong nuclear force diminishes rapidly with increasing separation and is negligible for separations greater than about $10^{-14}\ m$.



NUCLEAR STABILITY & RATIO OF NEUTRONS TO PROTONS

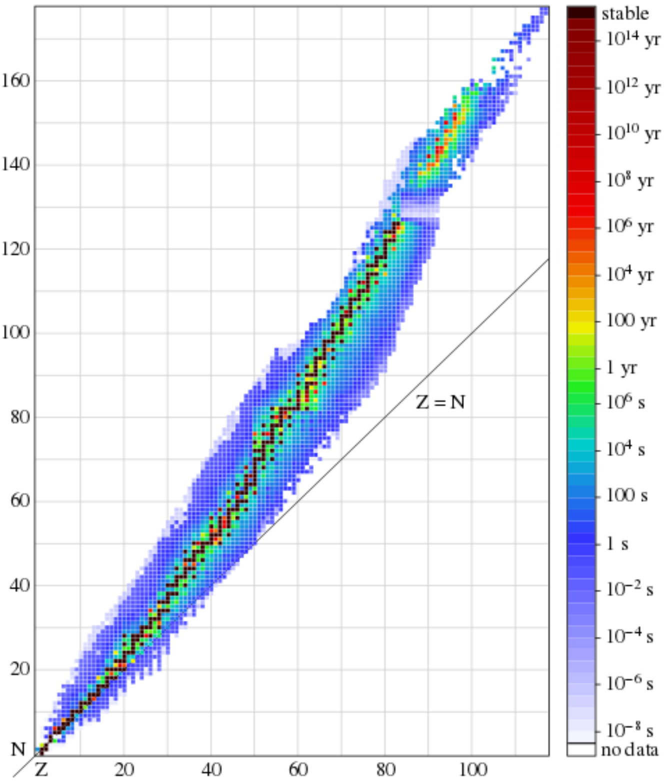
There are two main forces acting on the nucleons in the nucleus:

- The **electrostatic force** repels all protons
- The **strong nuclear force** that affects all nucleons depending on their separation.
- The **interplay** between these forces determines the stability of a nucleus.

Nuclear composition determines whether an isotope is radioactive of stable. A nucleus is unstable if:

- It falls outside the zone of stability (unstable neutron-to-proton ratio).
- The number of protons in the nucleus exceeds 82. Also, unstable if it has 126+ neutrons.
- There are no stable nuclei with Z > 82 (atomic number).

- For small atoms, stable nuclei have neutron-to-proton ratios near **1:1.**
- For large atoms, stable nuclei have neutron-to-proton ratios near **1.5:1.** This is because a larger number of neutrons is needed to stabilise the strong repulsion amongst the positively charged protons.
- Stability is favoured by **even numbers** of protons and even numbers of neutrons.



ENERGY IN NUCLEAR REACTIONS

OUTCOMES

- model and explain the process of nuclear fission, including the concepts of controlled and uncontrolled chain reactions, and account for the release of energy in the process (ACSPH033, ACSPH034)
- analyse relationships that represent conservation of mass-energy in spontaneous and artificial nuclear transmutations, including alpha decay, beta decay, nuclear fission and nuclear fusion (ACSPH032)
- account for the release of energy in the process of nuclear fusion (ACSPH035, ACSPH036)
- predict quantitatively the energy released in nuclear decays or transmutations, including nuclear fission and nuclear fusion, by applying: (ACSPH031, ACSPH035, ACSPH036)
  - the law of conservation of energy
  - mass defect
  - binding energy
  - Einstein's mass–energy equivalence relationship $E = mc^2$

ENERGY-MASS EQUIVALENCE

According to the Special Theory of Relativity mass is a form of energy, as expressed by: $E = mc^2$

Rest mass is not conserved in nuclear reactions

- If energy is **released** in the reaction, the **total energy** of the products will **decrease** and therefore so its mass
- If energy is **absorbed** in the reaction, the **total energy** of the products will **increase** and therefore so its mass

MEGA ELECTRONVOLTS (MEV)

- $1\,u = 1.661 \times 10^{-27}\,kg = 935.1\,MeV/c^2$
- $1\,eV = 1.602 \times 10^{-19}\,J$

TOTAL RELATIVISTIC MASS

The total relativistic mass of a particle is proportional to its **total energy**

$$m = \frac{E_{total}}{c^2}$$

$$m = \frac{1}{c^2}(m_o c^2 + K + U)$$

MASS DEFECT IN A NUCLEUS

The **mass of the atom** is **less** than the sum of the masses of its components (protons, neutrons, electrons).

The **mass of the nucleus** is **less** than the sum of the masses of its components (protons, neutrons)

The difference in mass is called the **mass defect** or **mass deficit**.

BINDING ENERGY

Binding energy is defined as the energy needed to **separate** the atom into its constituent parts.

- The mass defect is associated with the nucleus of the atom where the nucleons are packed into a small volume.
- Energy is released before a stable packing state is reached
- **Binding energy corresponds to the measured mass defect by Einstein equation.**

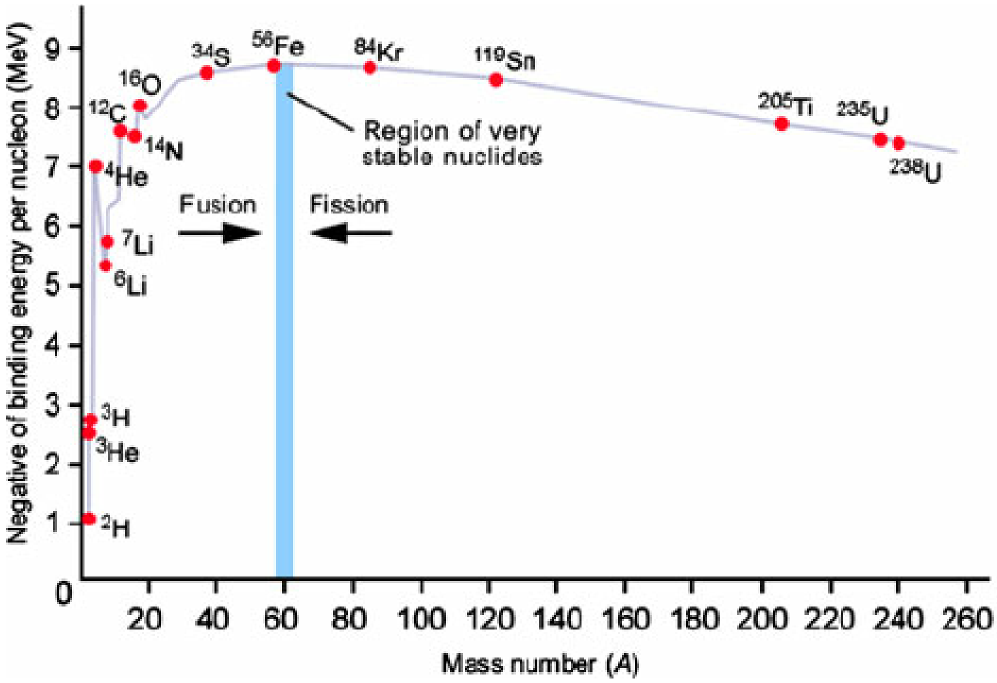
BINDING ENERGY PER NUCLEON

The binding energy per nucleon is related to the **stability of the nucleus.**

- The binding energy is released when nucleons combine to form the nucleus. By losing energy, a more stable state is obtained.
- A more useful measure for comparing stability is to look at the binding energy per nucleon.
- The binding energy per nucleon can be found by dividing the total binding energy by the number of nucleons in the nucleus.

$$Binding\ Energy\ per\ Nucleon = \frac{Binding\ Energy}{number\ of\ nucleons}$$

The **average binding energy per nucleon** indicates the **stability of a nucleus**, with larger values indicating higher stability.



Atoms whose **mass numbers** are close to **60** are **most stable** because they have the **highest binding energy per nucleon**.

- Two of the most stable isotopes are $^{56}_{26}Fe$ and $^{62}_{28}Ni$ with binding energy of 8.8 MeV/nucleon
- This suggests that the stability of the nucleus decreases as more nucleons are packed into the nucleus.

In general, a mass number:

- **< 60** will release energy by **disintegrating** through **fission reactions**
- **> 60** will release energy by **combining** through **fusion reactions**

ENERGY IN NUCLEAR REACTIONS

1) Consider each nucleus to be a unique entity with its own rest mass. If the products have less rest mass than the reactants, then some rest mass has been converted into other forms of energy (kinetic and light)
2) Consider each nucleus to be a unique arrangement of protons and neutrons, which have constant rest mass, but different nuclear potential energies in different nuclei. When nucleons drop to lower nuclear potential energies (i.e. greater stability in a nuclear reaction, the lost potential energy is converted into other forms of energy).

ENERGY RELEASED IN A NUCLEAR REACTION

In all nuclear reactions, rest mass is not conserved as it can be converted into energy. Also, some energy can be converted to mass.

If **rest mass is converted to energy**, energy will be released as the **kinetic energy of the products**.

$$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$$

If **energy is converted to rest mass**, energy must be **supplied** in order for the reaction to proceed.

$$^{4}_{2}He + {}^{4}_{2}He \rightarrow {}^{8}_{4}Be$$

In any nuclear reaction, the **total rest mass of the product is different to that of the reactants**. The difference is called **mass defect**.

$$Mass\ Defect = Mass\ of\ Reactants - Mass\ of\ Products$$

1 unified mass unit (u) = $1.661 \times 10^{-27} kg$

- Mass of the proton $m_p = 1.007276\ u = 1.673 \times 10^{-27}\ kg$
- Mass of the neutron $m_n = 1.008665\ u = 1.675 \times 10^{-27}\ kg$
- Mass of the electron $m_e = 0.000549\ u = 9.110 \times 10^{-31}\ kg$

1 unified mass unit converted into energy is $E = 1.661 \times 10^{-27} \times (3 \times 10^8)^2 = 931.5\ MeV$

NUCLEAR FISSION REACTIONS

Binding energy per nucleon decreases for nuclei with a mass greater than 60.

- This means **large nuclei** with masses over 60 can **release energy** if they are **broken into smaller pieces**
- This is called a **fission reaction**. Fission is the process by which a nucleus splits into two lighter (daughter) nuclei.

MECHANISMS FOR NUCLEAR FISSION REACTIONS

Nuclear fission can occur spontaneously in unstable nuclei as an uncommon form of decay or can be triggered by a bombarding particle.

The fission of ${}^{235}_{92}U$ by slow, low energy neutrons is an example of a fission reaction. The process of nuclear fission of the uranium nucleus proceeds as follows:

1) The ${}^{235}_{92}U$ nucleus **captures** a low energy neutron
2) The capture results in the formation of ${}^{236}_{92}U^*$, and the excess energy of the nucleus causes it to undergo violent oscillations.
3) The ${}^{236}_{92}U^*$nucleus becomes highly **distorted** and the force of repulsion between protons in the two halves of the dumbbell shape tends to increase the distortion. The strong force weakens due to the increasing distance.
4) The nucleus **splits** into two fragments, emitting several neutrons in the process, as well as energy.

In the fission of uranium-235, there are about 90 different daughter nuclei that can be formed as fission fragments:

- The process also results in the production of several neutrons; typically, 2 or 3
- An example reaction for uranium-235 is

$$ {}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{142}_{58}Ba + {}^{91}_{38}Kr + 3{}^{1}_{0}n $$

FISSION CHAIN REACTIONS

A chain reaction refers to a situation where a reaction is triggered by a previous reaction. A nuclear fission chain reaction occurs when a fission reaction is triggered by a neutron, and then releases more neutrons. The released neutrons can be absorbed by other nuclei, triggering more fission reactions.

- If **only one neutron per event** is allowed to **cause another fission**, the reaction is **controlled and self-sustaining**
- If **multiple neutrons are allowed to cause further fission**, the reaction is **uncontrolled** and can result in an explosion.

NUCLEAR FISSION REACTORS

A nuclear reactor is a system designed to maintain a **controlled self-sustained chain reaction** and to release nuclear energy at a controlled rate.

- Self-sustained chain reactions do not require continual energy input from external sources so allow for the easy continual production of energy

Nuclear reactors are classified as:

- **Thermal reactors:** the neutrons producing the fission have energies comparable to gas particles at room temperature. Most commercial reactors are thermal reactors.
- **Fast reactors:** The neutrons producing the fission have high energies

REACTOR COMPONENTS

The essential components of a thermal fission reaction are;

1) A core of fuel
2) A neutron moderator
3) A coolant
4) Control rods
5) Radiation shielding

FUEL

The fuel must contain **fissionable** material. Natural uranium is composed of three isotopes:

- U-238 (99.283%)
- U-235 (0.711%)
- U-234 (0.006%)

MODERATORS

The **moderators** have the job of **slowing down the neutrons** to make them available for reaction with U-235 and to decrease their chances of being captured U-238.

The moderator can be made of **graphite** or **water**, and surrounds each fuel rod. **Most modern reactors use water as the moderator**.

The neutrons released in fission events are very energetic, with kinetic energies of about 1-2 MeV.

U-235 absorbs neutrons best when they are slow, or thermal, with energies of around 0.04 eV. When U-235 absorbs a neutron, it undergoes fission, releasing neutrons.

U-239 absorbs fast neutrons better than U-235. When it absorbs a neutron, it is mostly likely to capture it and undergo beta decay, **not releasing neutrons**.

In order to understand how neutrons are slowed, consider a collision between a light object and a very massive one:

- In such an event, the light object rebounds from the collisions with most of its original kinetic energy.
- However, if the collision is between objects whose masses are nearly the same, the incoming projectile will transfer a large percentage of its kinetic energy to the target.
- During an elastic collision between two particles, the maximum kinetic energy is transferred from one particle to the other when they have the same mass.
- The neutron can undergo multiple collisions until it reaches thermal energies.

The way a **neutron** interacts with the above process depends on the **nature** of the material with which the neutron interacts.

- In moderators, **elastic collisions are dominant**. Moderators thus slow down (or moderate) the originally energetic neutrons very effectively **without absorbing them**.

COOLANT

The **kinetic energy** of neutrons is converted to **heat** during **collisions** with the moderator and coolant.

- The coolant transfers heat away from the reactor core to where it can do useful work
- The coolant may be air, helium, heavy water, liquid sodium or some organic compound.
- Since the coolant has been inside the reactor core, it may be radioactive and cannot be allowed to escape.

CONTROL RODS

The fission chain reaction depends upon the number of neutrons available for absorption.

The chain reaction can eb controlled by inserting **control rods** into the reactor core. It controls the power level by absorbing neutrons (without emitting any) since the fission chain reaction depends upon the number of neutrons available.

The rods are usually made of **steel containing boron or cadmium**.

RADIATION SHIELD

Enormous amounts of gamma radiation are released within the reactor core. To ensure the safety of workers, two type of shielding are used.

- A shield of **graphite and lead** protects the walls from radiation damage and to reflect escaping neutrons back into the core.
- A **biological shield** made of high density concrete many centimetres thick absorbs radiation as much as possible.

NUCLEAR FUSION REACTIONS

Binding energy per nucleon decreases as mass number decreases for nuclei with a mass number lower than 60.

- This means **small nuclei** with masses below 60 can release energy if they are **joined together to form larger nuclei**.
- This is called a **fusion reaction**. **Fusion** is the process by which **lighter nuclei are joined together to form a larger nucleus**.

It is extremely difficult to maintain a nuclear fusion reaction on earth as the nuclei are positively charged and will exert electrostatic forces causing them to be repelled from one another. The nuclei must be travelling towards each other at high enough speeds such that the kinetic energy can overcome the repulsive forces. In order for the nuclei to overcome the energy barrier, the nuclei must be in an environment that is hundreds of millions of degrees. This enormous amount of energy enables it to overcome the energy barrier and fuse together. Currently, there is no way to sustain such a higher temperature reaction on Earth.

STANDARD MODEL OF MATTER

OUTCOMES

- analyse the evidence that suggests:
  - that protons and neutrons are not fundamental particles
  - the existence of subatomic particles other than protons, neutrons and electrons
- investigate the Standard Model of matter, including:
  - quarks, and the quark composition hadrons
  - leptons
  - fundamental forces (ACSPH141, ACSPH142)
- investigate the operation and role of particle accelerators in obtaining evidence that tests and/or validates aspects of theories, including the Standard Model of matter (ACSPH120, ACSPH121, ACSPH122, ACSPH146)

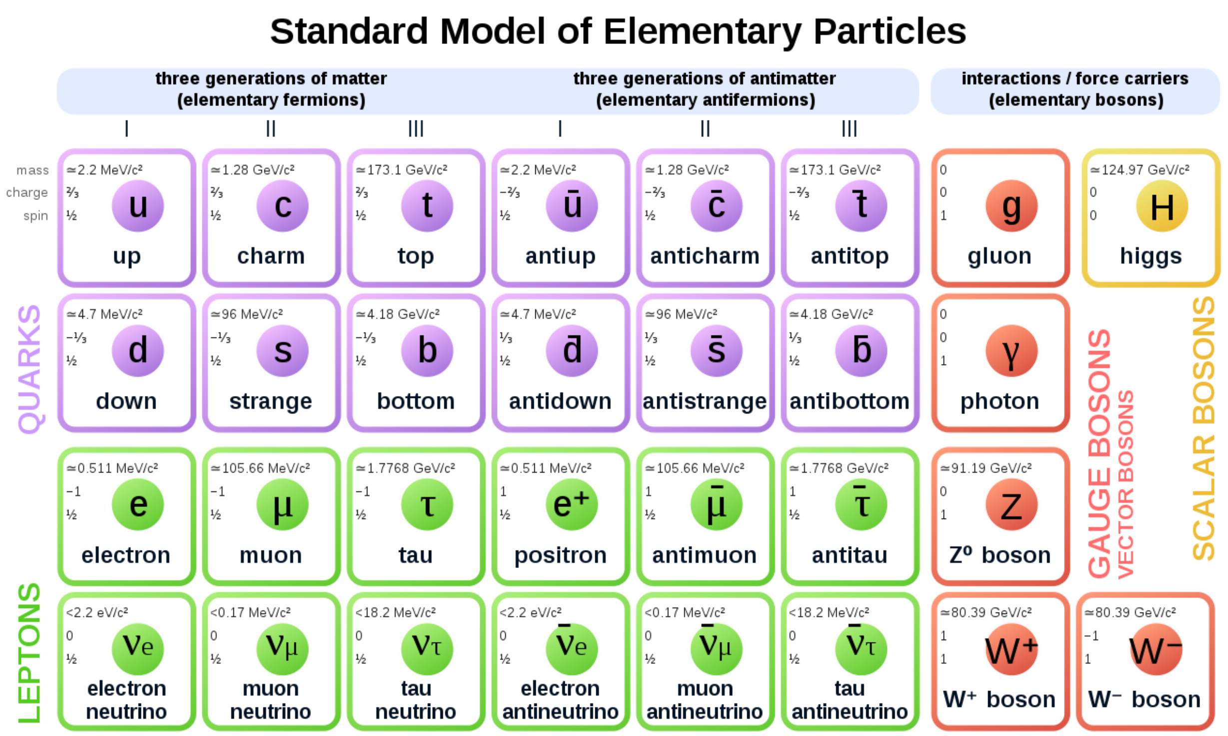
STANDARD MODEL OF MATTER

Physicists have developed a theory known as the **Standard Model** that explains our current understanding of the nature of matter:

- What are the components of matter?
- How do the components of matter interact with each other?

According to the Standard Model, all the particles in the universe can be grouped into just three "families" of elementary particles:

- 6 quarks, which combine to form nucleons and other particles
- 6 leptons, including the electron and neutrinos
- 5 force carrier particles, including the photon
- Their anti-particles



All the discovered **matter particles** are **leptons** or **composites of quarks** and they **interact by exchanging force carrier particles**.

ANTIMATTER

For each kind of **matter** particle, there is a corresponding **antimatter** particle.

- Each **quark** and **lepton** have a corresponding **antiquark** and **antilepton**.
- The photon, Z, gluons and Higgs bosons are their **own antiparticles**
- The W^+ and W^- bosons are each other's antiparticles

Antiparticles look and behave just like their corresponding matter particles, they have the **same mass** but they have the **opposite charge**.

When a matter particle and its corresponding antiparticle meet, they **annihilate** into pure energy.

- An electron and positron will annihilate to produce two photons (gamma rays) of the same energy. This is called **pair annihilation**. All the mass of the two particles is **converted** into energy according to $E = mc^2$.

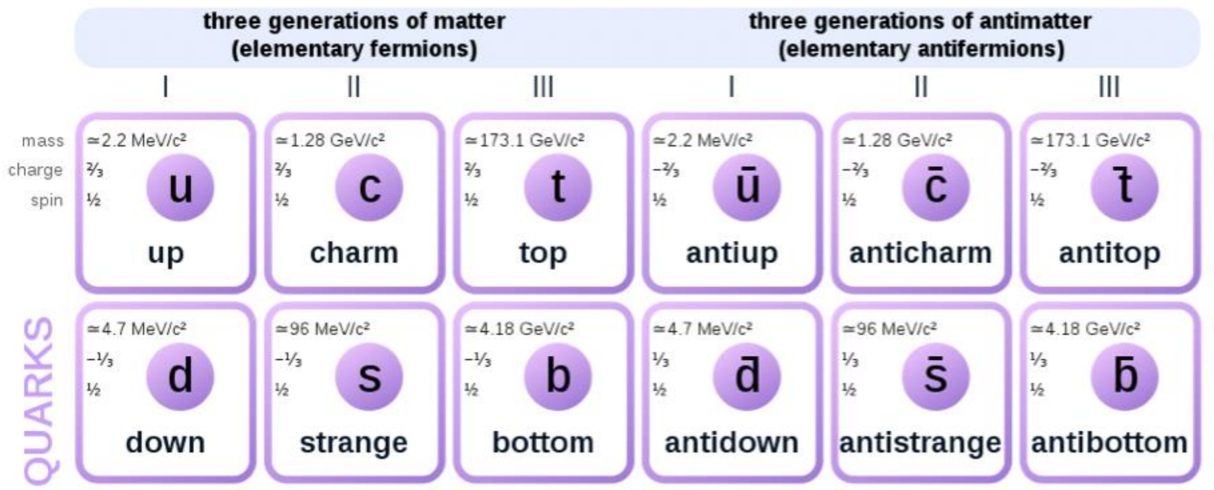
FERMIONS

Fermions are a fundamental particle. It is divided into 2 groups called **quarks** and **leptons**.

QUARKS

Protons and neutrons are not fundamental – they are made up of even smaller particles called quarks. Quarks are usually found in combinations, in composite particles like protons and neutrons.

Quarks experience all 4 fundamental forces.



Quarks have fractional electric charges. The up, charm, and top have a charge of $+\frac{2}{3}e$, and the down, strange, and bottom have a charge of $-\frac{1}{3}e$.

For each of these quarks, there is a corresponding **antiquark**, which has the same mass but opposite charge.

Baryon are particles that have a baryon number of 1 for normal matter or -1 for antimatter. They include **protons** and **neutrons** which consist of 3 quarks.]

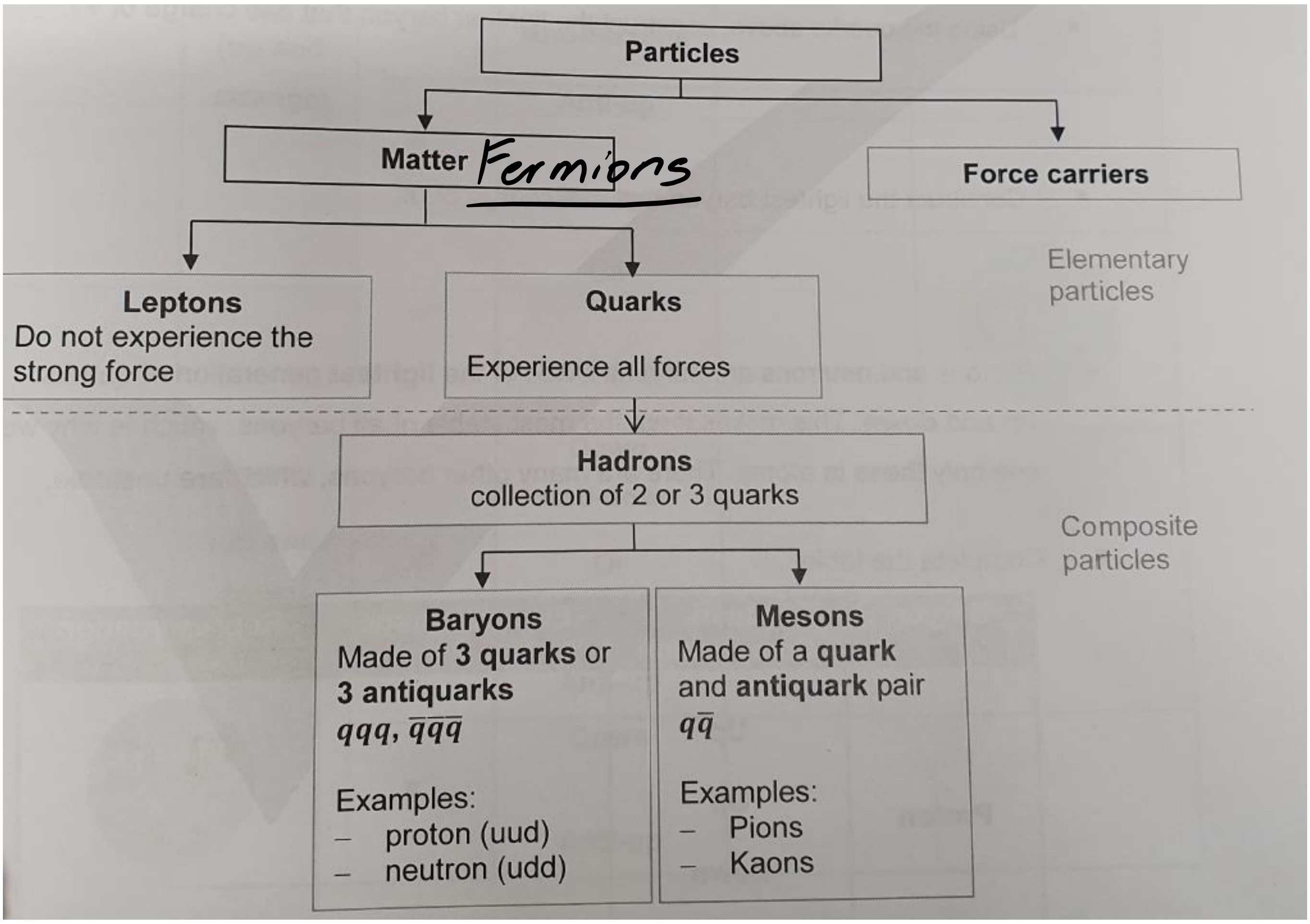
HADRONS, BARYONS AND MESONS

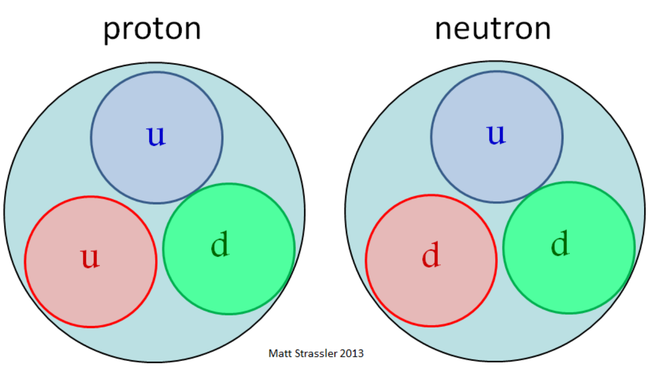
Quarks exist in groups with other quarks. They combine to form composite particles called **hadrons** which are **held together by the strong force**.

- The fractional electric charges of quarks always combine such that the hadron **has a net integer electric charge**.

There are two common classes of hadrons:

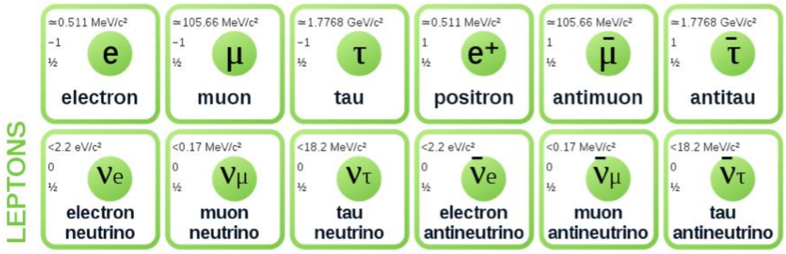
- **Baryons** are any hadrons which are made out of three quarks or three antiquarks.
- **Mesons** are any hadrons which are made of one quark and one antiquark.





LEPTONS

Leptons are the other type of matter particles.



Leptons **do not experience the strong force**, so they **cannot be held in a nucleus.** There are 6 leptons, three of which have electrical charges and three do not. They appear to be point-like particles without internal structure (fundamental).

Leptons are divided into 3 lepton pairs:

- The electron and its neutrino
- The muon and its neutrino
- The tau and its neutrino

The best-known lepton is the electron.

- The other two charge leptons are muon and the tau, which are like charged like electrons but have a lot more mass. They are unstable.
- There are three types of neutrinos. They have no electrical charge, very little mass and are extremely hard to detect.

MESONS

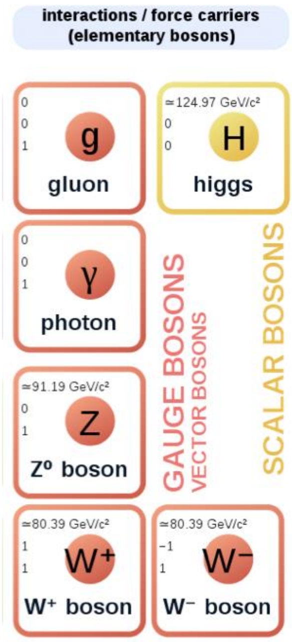
All mesons consist of a **quark** and an **antiquark** (matter and antimatter) and therefore all decay quickly.

There are many different mesons arising from combination of quarks and antiquarks. There are two families of mesons: **pions** and **kaons**.

Mesons	Symbol	Quark Composition	Electric Charge	Representation
Kaons (up and strange)	K^+	Up & Anti-Strange	$\frac{2}{3}+\frac{1}{3}=1$	$u-\bar{s}$
	K^-	Anti-Up & Strange	$-\frac{2}{3}+\frac{1}{3}=1$	$\bar{u}-s$
Pions (up and down)	π^+	Up & Anti-Down	$\frac{2}{3}+\frac{1}{3}=1$	$u-\bar{d}$
	π^0	Down & Anti-Down or up & Anti-Up	0	$d-\bar{d}\ or\ u-\bar{u}$
	π^-	Down & Anti-Up	$-\frac{1}{3}-\frac{2}{3}=-1$	$d-\bar{u}$

FORCE CARRIER PARTICLES

There are several kinds of forces that particles can exert on one another. These forces can cause one particle to attract or repel another particle, and can even cause a particle to transform into a different type of particle.



All interactions between particles can be explained in terms of **four fundamental forces**.

- Electromagnetic force (photon)
- Strong nuclear force (gluon)
- Weak nuclear force (W and Z bosons)
- Gravitational force (not explained by the standard model of matter)

The **exchange** of bosons results in the forces. Bosons are often called force-carrying, force-mediating or exchange particles. Analogies for the exchange of bosons are:

- Two inline skaters stand stationary and then began passing footballs back and forth to each other. As they do this, they will begin to move away from each other. This is due to the conversation of momentum each time they throw and catch the ball. This situation could be likened to two particles experiencing a repulsive force.
- Two inline skaters now exchange the footballs by trying to grab them out of each others' hands. They will exert a force of attraction on each other. This would cause them to move together and can be likened to two particles experiencing an attractive force.

ELECTROMAGNETISM: PHOTONS

All electromagnetism is due to (virtual) **photons** being transferred between charged particles.

Feynman diagram for the electromagnetic force between two charges.

STRONG FORCE: GLUONS

The strong nuclear force is a force between **quarks** carried by particles called **gluons**.

- This force glues quarks together to form hadrons and mesons

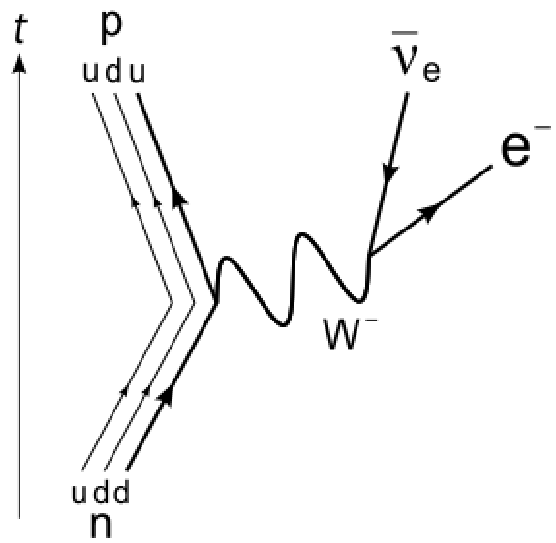
Gluons can only have an effect over a very short range. Although the strong forces interacts between quarks in the same nucleon, it can also interact between different nucleons, so it can hold together the nucleus.

WEAK FORCE: W AND Z BOSONS

The weak force has three bosons associated with it: W^+, W^- and Z^0. All these bosons are relatively **heavy**, so they decay quickly and thus the weak force has **short range**.

The weak force is the only force that can change the **flavour** (type) of a quark or lepton. It allows quarks and leptons to interact. The weak force is the only force that can change the **charge** of a particle, because $W^+, W^-$are the only **charged** force carriers.

The most common example of a weak interaction is **beta decay**. Beta-minus decay is due to the **weak force** turning **a down quark** into an **up quark**, hence a **neutron into a proton.**



- The W^- boson carries away the $-e$ of charge from the down quark and the down quark turns into an up quark.
- The W^- then decays into an electron and electron antineutrino which are the radiation emitted by beta-minus decay

SUMMARY OF FORCES

Force	Force Carrier	Properties
Electromagnetic	Photons	- Range of effect is infinite since it has no rest mass - Experience by everything with charge (quarks, leptons, $W^{\pm}$) - Holds atoms and molecules together through the attraction between the nucleus and the electrons, and the attraction between atoms in a molecule.
Strong	Gluons	- Short range (1fm) - Experience by quarks and gluons - Holds hadrons together, and holds the nucleus together
Weak	W^+, W^-, Z^0	- Short range - Experience by quarks and leptons - Can change the charge of particles - Responsible for beta decay and other decay reactions

SUMMARY OF THE STANDARD MODEL OF MATTER

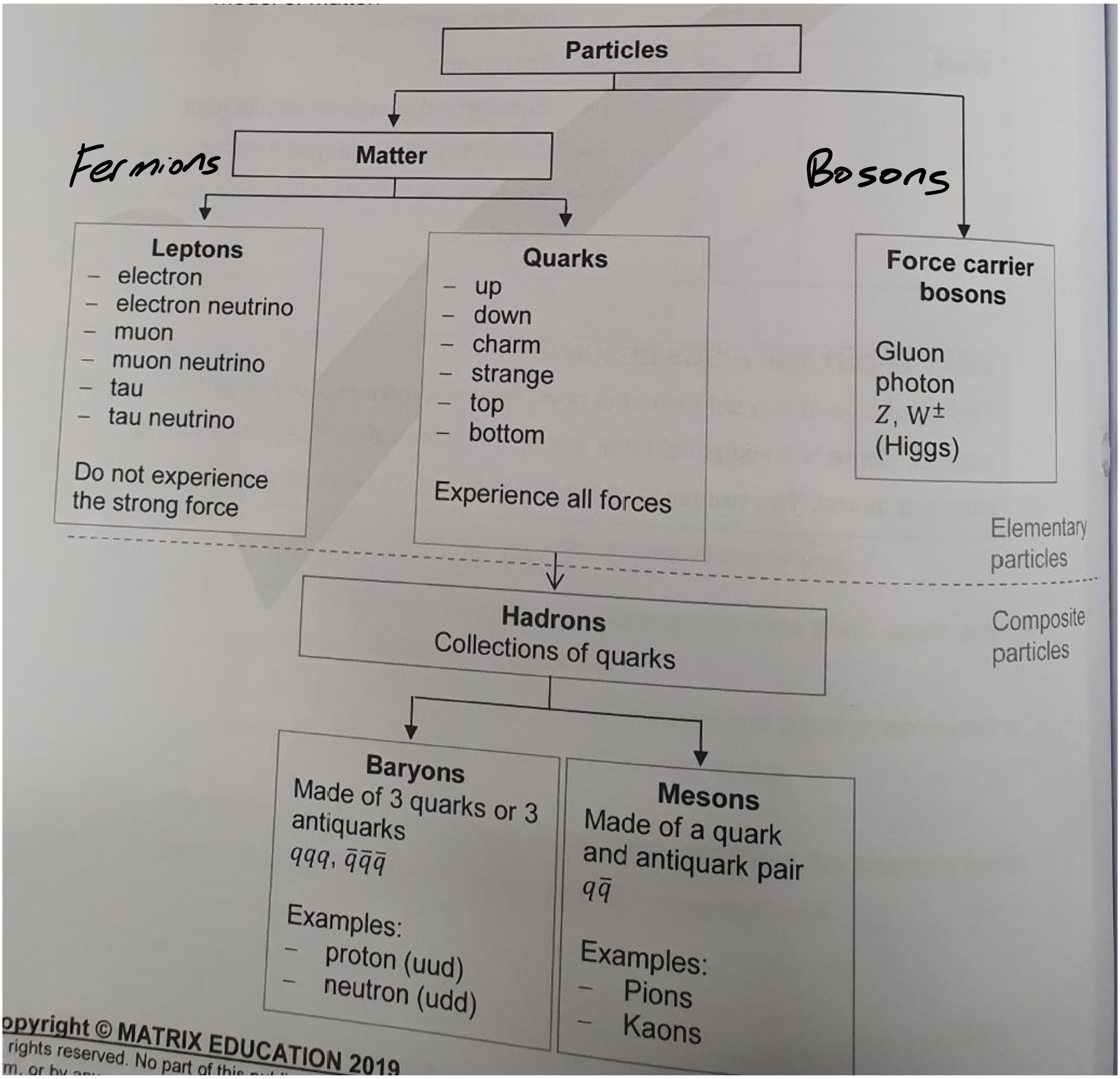
Quarks experience all forces

- A Baryon is a composite particle made up of 3 quarks
- A meson is a quark-antiquark pair
- Protons and neutrons are made up of down and up quarks

Leptons, including electrons and neutrinos, do not experience the strong nuclear force, but experience the weak nuclear force.

- Electrons are the only lepton that exist in ordinary matter

Force carrier bosons carry electromagnetic, weak and strong forces.



PARTICLE ACCELEARTORS

Most of the Standard model was verified using **particle accelerators**.

- Particles are accelerated to near the speed of light, giving them much higher energy than they would ordinarily have.
- The high-energy particles are made to collide
- The kinetic energy is converted to rest mass, creating new particles that can be studied

Particle accelerators are essential probes in discovering the structure of matter, collecting evidence leading to the Standard model, and then experimentally supporting it.

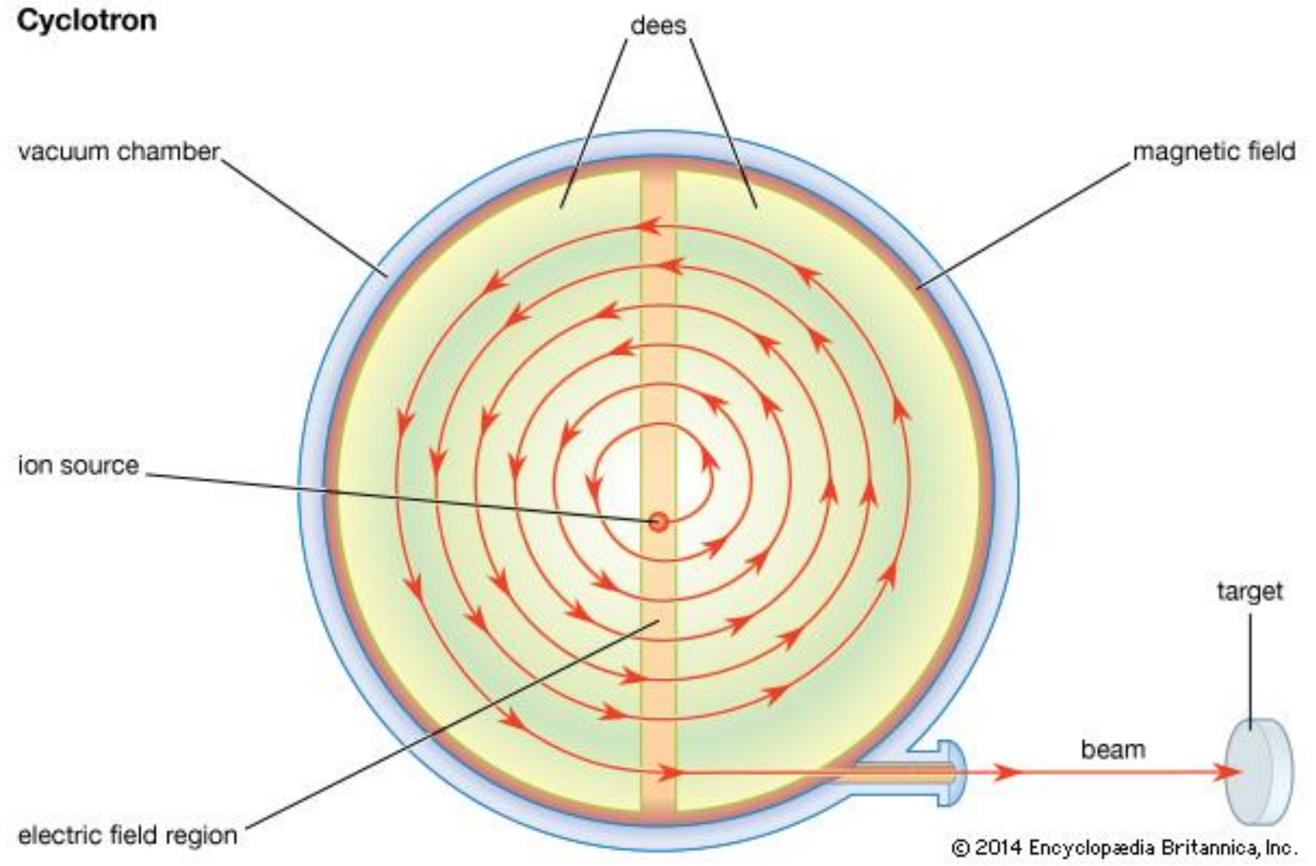
LINEAR ACCELEARTORS

The largest linear accelerator is the **Stanford Linear Accelerator (SLAC)**. Experiments conducted first discovered that protons are not fundamental particles. This was a turning point for our understanding of matter and the Standard Model.

The experiment that proved protons are not fundamental particles is called **deep inelastic scattering**.

- Electrons and protons are fired at each other at high speed
- At lower energies, the electron and proton elastically scatter
- At high energies, the **proton absorb some energy from the electron which makes the collision inelastic.** The proton is then able to **emit a quark**.
- If the proton is able to **absorb energy** and **emit a particle**, it cannot be a fundamental particle.

THE LAWRENCE CYCLOTRON



- A small tube leads a stream of hydrogen gas to the centre of the box where a spark strips away the electron form each atom.
- An alternating electric field between the gap accelerates the proton when passing across the gap.
- The magnetic field steers the proton into a circular trajectory. The magnetic field direction is perpendicular to the plane of the circle.
- As the protons accelerate in a circular trajectory, they pass cross the gap between the two copper 'D's.
- As the protons gain energy, the radius of their orbit increases and they gradually spiral out to the limit of the magnetic poles area. Air is removed from the entire region so that collisions with air molecules do not interfere with the accelerator.

SYNCHROTRON

The synchrotron accelerator developed from the cyclotron.

- Like the cyclotron, it uses a magnetic field to store the charges for multiple passes through an accelerating potential difference.
- However, in a synchrotron, the particles travel in a circular loop rather than a spiral. This means that the synchrotron can just be a ring instead of an entire disc.
- The magnetic field can be concentrated on just the position of the particle, rather than the whole area, and it is easier to make a larger ring than a large disc.

In a synchrotron, the **path travelled is a circle**, however the **frequency** of the electric field and **strength** of the magnetic field has to change depending on the **velocity** of the particle.

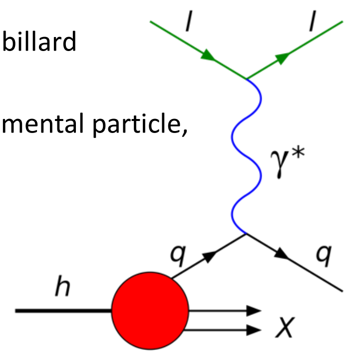
- The electric and magnetic fields have to be **synchronised** to the position and velocity of the particles
- This means the synchrotron needs much more advance control than the cyclotron, and is computer controlled.

LINEAR ACCELEARTORS

The largest linear accelerator is the **Stanford Linear Accelerator (SLAC)**. Experiments conducted at SLAC in 1969 first discovered that protons are not fundamental particles. This was the turning point in the understanding of the Standard Model of Matter.

The experiment that proved protons are not fundamental particles is called **deep inelastic scattering**

- Electrons and protons are fired at each other at high speeds.
- At lower energies, the electron and protons elastically scatter.
- At high energies, the **proton absorbs some energy from the electron** which makes the **collision inelastic**. The proton is then able to **emit a quark**.
- Scattered if as they were 3 dense positive charges in a proton. It was not like billard balls.
- f the proton is able to absorb energy and emit a particle, it cannot be a fundamental particle, leading to the discovery of quarks.



LARGE HADRON COLLIDER

The Large Hadron Collider is an enormous collider. It uses a **linear accelerator** to initially accelerate the particles, then a series of **synchrotrons** to accelerate the particles in stages.

It was built after previous results from accelerators hinted at the existence of the Higgs boson, the particle that is hypothesized to give particles their mass. In 2012, the LHC detected a particle that behaved like a Higgs boson.

The Large Hadron Collider offers cosmologist an insight into some of the detail of the conditions present shortly after the Big Band, and the opportunity to study the behaviour and interactions of the various matter and anti-matter particles.

THE BIG BANG THEORY

OUTCOMES

- investigate the processes that led to the transformation of radiation into matter that followed the 'Big Bang'
- investigate the evidence that led to the discovery of the expansion of the Universe by Hubble (ACSPH138)

EDWIN HUBBLE

The discovery that the **universe is expanding** was made in 1929 by **Edwin Hubble**. He proved that the universe was indeed **not static** and dynamic.

He proved that the universe is expanding by showing that **galaxies moved away from the Earth at a velocity that was proportional to their distance**.

In 1929, Hubble used Slipher's red shift measurements and combined them with distance measures made using Cepheid variables.

- He showed that the further away a galaxy was relative to the Earth, the greater the velocity of the galaxy.
- This could only be explained if the universe was expanding

Hubble's law is the **speed of recession** is proportional to the **distance away**.

$$v = H_0 d$$

Where:

- v: The speed of recession km/s
- H_0: Hubble's constant
- d: distance away (mega parsecs, Mpc)

COSMIC REDSHIFT

Cosmic redshift refers to the redshift of the observed wavelength of radiation due to the expansion of space.

- Light emitted by a faraway object like a distance galaxy **will take a long time to reach Earth**.
- Over that time, **space will expand**.
- The **wavelength of light will be stretched by the expansion of space**. Thus the observed wavelength will be **longer** when it reaches Earth, due to cosmic redshift.

There are **differences** between the cosmic redshift and the Doppler Effect.

- The Doppler Effect depends on the relative velocity of the source and observer at the time the light is emitted, and requires the source or observer or both to be moving in space.
- The cosmic redshift occurs after the light has been emitted due to the expansion of space. It does not require the source or observer to be moving in space.

However, Doppler Effect and cosmic redshift will be used interchangeably since the **expansion of space results in relative velocity**.

LUMINOSITY

The luminosity L is the **power emitted by a star** and is measured in watts.

Stars are assumed to be black bodies. The amount of power radiated by a black body is given by the Stefan-Boltzmann law:

$$L = \sigma \mathrm{A} \mathrm{T}^4$$

Where:

- L: Luminosity (J/s or watts)
- σ: Boltzmann constant
- A: Surface area of the star
- T: Temperature of the star

BRIGHTNESS

The brightness of a star refers to how bright it appears from Earth. It is a measure of intensity and has the units watts per meter squared.

$$B = \frac{L}{4\pi r^2}$$

READY STATE THEORY

- The universe is infinite. The outer stars would never reach infinity and would move away from us forever.
- Matter is constantly being created to keep the density of the universe constant.

BIG BANG THEORY

The main idea in the Big Bang Theory is that everything was created from 1 infinitely dense point called a singularity.

OVERVIEW OF THE BIG BANG THEORY

At the Big Bang, 13.7 billion years ago, a singularity **expanded**, creating **space, time, energy and matter**. The Big Bang is the birth of the universe at the beginning of time.

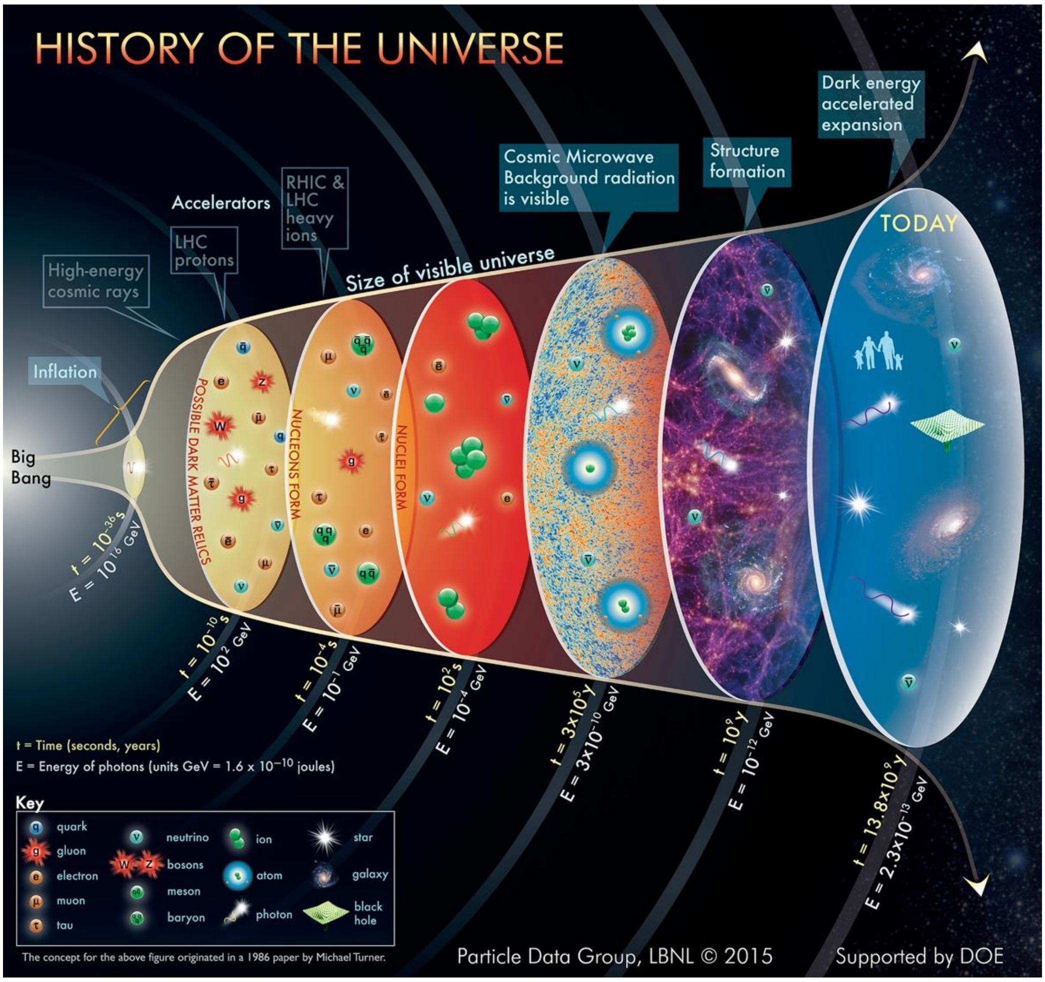
The **first phase** of evolution was driven by **expansion and cooling**

- Expansion led to cooling as the energy of the universe was spread over an increasingly larger volume.
- Expansion and cooling allowed **energy to be converted into matter**, in the form of quarks, leptons and force carrier particles. Eventually these decayed to form protons, neutrons, electrons and photons, then natural gas (mostly hydrogen and helium)
- Expansion and cooling dominated the evolution of the universe for the first 380,000 years.

The **second phase** of evolution was driven by **gravity**

- Gravity led to the gas in the universe collapsing into giant clouds and eventually forming the **stars and galaxies we see today**.
- The first stars and galaxies formed 200-400 million years after the Big Bang.

Process/State of the Universe	Time	Temp (K)
The Big Bang: the singularity corresponds to very small, hot, dense universe where all energy is confined to a small volume	0	∞
Inflation: The universe expand **rapidly** and **cools** as energy is spread over a larger volume. **Energy is converted into matter** forming unstable fundamental particles.	$10^{-32}\ s$	$> 10^{27}$
Post Inflation: The universe **expands** and **cools** at a **slower rate**. The unstable fundamental particles interact and decay eventually resulting in only stable particles: protons, neutrons and photons	$10^{-6}\ s$	$> 10^{13}$
Nucleosynthesis: Further cooling allows **protons** and **neutrons** to react and form **helium** and small amounts of lighter elements.	$0.01\ s\ to\ 3\ mins$	$> 10^{8}$
Recombination: Further cooling allows nuclei and electrons to form **neutral atoms**. **Decoupling between matter and radiation occurs at this time**, as the universe changes from being ionised and opaque to neutral and transparent.	$380{,}000\ yr$	4000
Reionisation: After a long period of accretion the **first stars** and **galaxies** form.	$400 \times 10^{6}\ to\ 1 \times 10^{9}\ yr$	$60 - 20$
Present day: The universe is filled with galaxies and stars at large distances from one another	$13.7 \times 10^{9}\ yr$	2.73



EVIDENCE FOR THE BIG BANG

There are **three** main pieces of evidence to support the Big Bang Theory:

1) The expansion of the universe
2) The abundance of light elements in the universe
3) The Cosmic Microwave Background radiation

ABUNDANCE OF LIGHT ELEMENTS

According to the Big Bang Theory, energy was converted into matter eventually forming protons, neutrons and electrons.

- Hydrogen in the universe was formed at this stage
- At high temperatures, hydrogen can undergo fusion reactions to form helium, and other lighter elements
- The formation of some light elements took place during the period of nucleosynthesis in the first 3 minutes after the Big Bang.

The Big Bang Theory predicts the amount of each element that should have been created, given how the temperature changed after the Big Bang due to expansion and cooling.

THE COSMIC MICROWAVE BACKGROUND RADIATION

In its first 380,000 years the universe was filled with charged particles.

- First it was filled with quarks, leptons and force carrier particles.
- After cooling down further, it became filled with **free protons, electrons** and **photons**.

Charged particles can absorb and emit electromagnetic radiation, meaning that during this time **radiation** (photons) was **coupled to matter**.

- Photons were being constantly absorbed and emitted by different particles and were not able to travel through space. Hence the universe was **opaque**.
- Radiation and matter were in equilibrium: the radiation was the black body radiation emitted by matter.

After 380,000 years after the Big Bang **recombination** occurred, meaning that **radiation recoupled from matter**:

- Protons and heavier nuclei combined with electrons to form **neutral gas**
- Neutral gas does not interact with radiation strongly.
- Hence the universe became **transparent**
- Radiation was not interacting with matter (not being absorbed or emitted) and would just travel through space forever.

In 1948, conclusions about Big Bang Theory model were made:

- There should be an **'afterglow'** of the Big Bang: the radiation that existed at recombination which was free to travel through space should still be travelling through space now and surround us everywhere.
- The wavelength of the radiation should be longer due to the expansion of space. They estimated that the wavelength should be in the microwave spectrum and called it the **cosmic microwave background radiation**.
- It was discovered in 1963 the blackbody temperature of the cosmic microwave background radiation is 2.7 K

ACCRETION OF GALAXIES AND STARS

ROLE OF GRAVITY

After recombination the universe was filled with neutral gas.

- Further expansion and cooling did not change the contents of the universe: it simply resulted in colder gas.
- As the gas cooled the kinetic energy (and speed) of the particles reduced.
- Once the gas particles were moving slowly enough, gravitational forces between them (although every weak) become important

In order to have a **net gravitational force** in this stage of the universe when it was filled with gas, an **uneven distribution of matter** is required.

ACCRETION

Accretion is the process by which denser objects attract nearby matter due to their gravity. Accretion is responsible for the formation of some structures in the universe, including stars and galaxies from the gravitational collapse of giant gas clouds.

1) Uneven distribution of matter
2) Variations in gravitational force strength
3) Gathering of mass: mass is attracted to denser regions by stronger gravity
4) Formation of giant gas clouds
5) Uneven distribution of matter inside the gas clouds
6) Fragmentation into smaller denser clusters
7) Gravitational collapse into denser clumps
8) Temperature increases
9) Stars are formed

The increased temperature in stars led to hydrogen gas in stars becoming ionised, hence the formation of first stars is referred to as **reionisation**.

An **uneven cloud** has areas of **higher density** that will result in a greater **gravitational force**. This leads to increasing density of these regions and results in the gathering of more matter.

Accretion led to the universe that we see today:

- Collapse of the gas from recombination into giant clouds formed galaxies
- The collapse of a gas within the giant clouds formed stars
- There is little material in between galaxies and in between stars

Accretion was a very slow process due to the weak nature of gravity and to the small differences in density, which resulted in a weak gravitational force.

- The time between recombination and reionisation was around 350 million to billion years. This period is called the "dark ages" as the cosmic background had red-shifted to the infrared, and stars had not yet formed, meaning there was no visible light in the universe.

ASTROPHYSICS

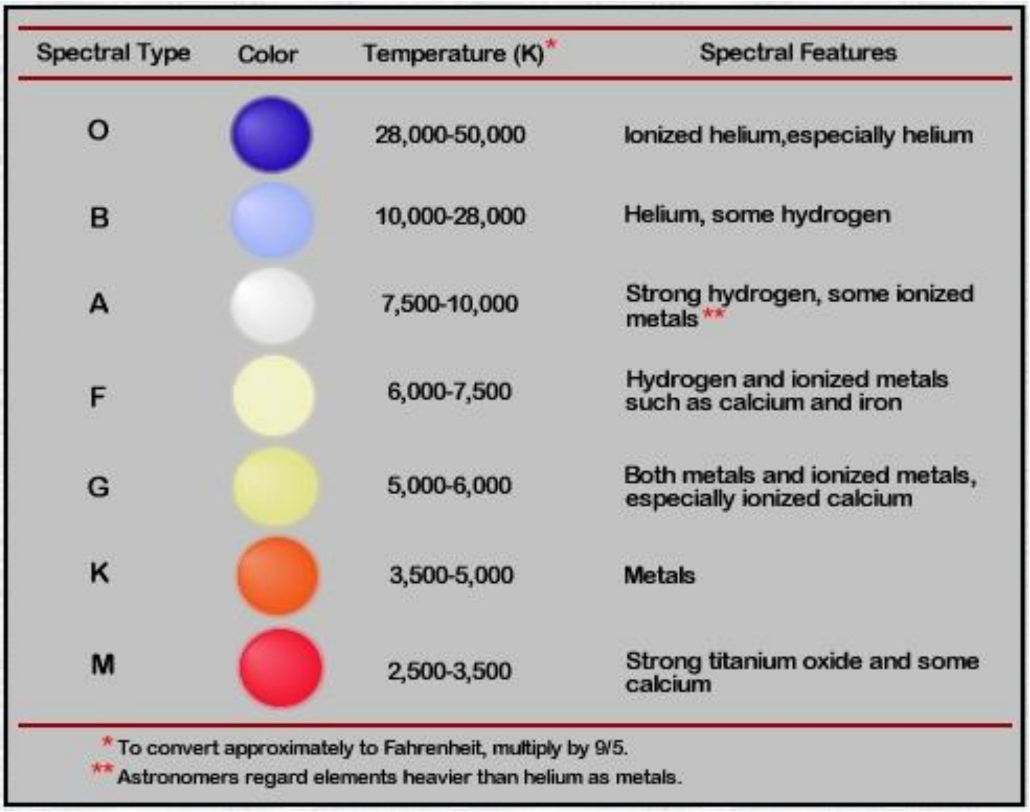
OUTCOMES

- analyse and apply Einstein's description of the equivalence of energy and mass and relate this to the nuclear reactions that occur in stars (ACSPH031)
- account for the production of emission and absorption spectra and compare these with a continuous black body spectrum (ACSPH137)
- investigate the key features of stellar spectra and describe how these are used to classify stars
- investigate the Hertzsprung–Russell diagram and how it can be used to determine the following about a star:
  - characteristics and evolutionary stage
  - surface temperature
  - colour
  - luminosity
- investigate the types of nucleosynthesis reactions involved in Main Sequence and Post-Main Sequence stars, including but not limited to:
  - proton–proton chain
  - CNO (carbon–nitrogen–oxygen) cycle

SPECTAL CLASSES

When scientists studied the spectra of stars, they found similarities between them that allowed them to group stars into categories called **spectral classes**.

- "Oh be a fine girl kiss me" OBAFGKM
- Each spectral class also corresponds to a **range of surface temperatures** (and **colours**).



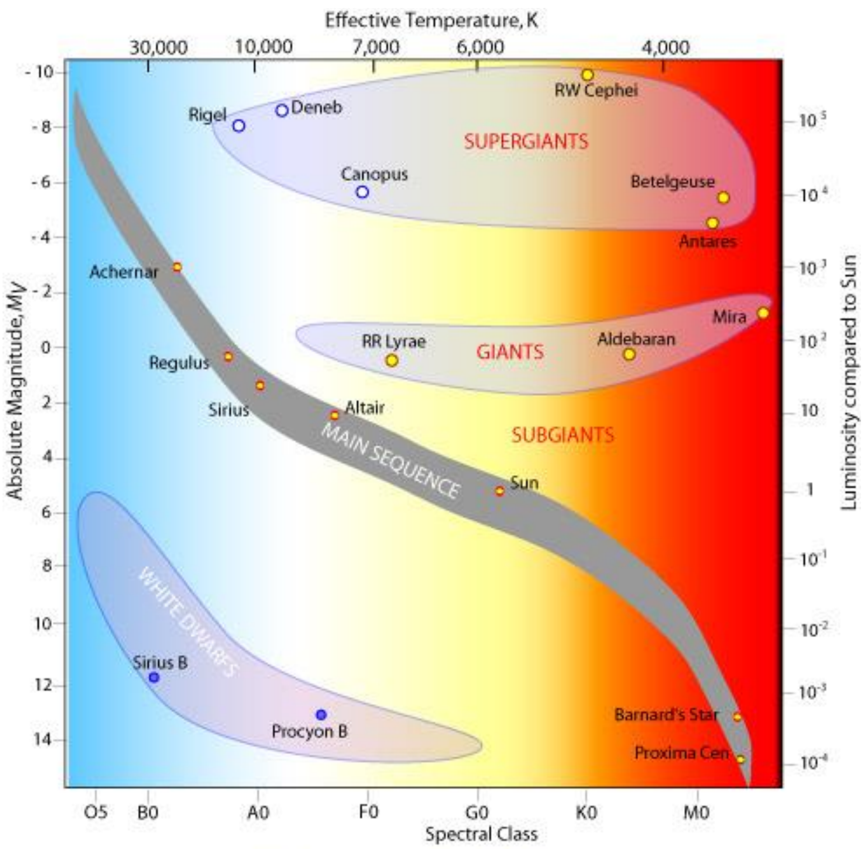
Spectral Type	Color	Temperature (K)*	Spectral Features
O		28,000-50,000	Ionized helium,especially helium
B		10,000-28,000	Helium, some hydrogen
A		7,500-10,000	Strong hydrogen, some ionized metals**
F		6,000-7,500	Hydrogen and ionized metals such as calcium and iron
G		5,000-6,000	Both metals and ionized metals, especially ionized calcium
K		3,500-5,000	Metals
M		2,500-3,500	Strong titanium oxide and some calcium

*To convert approximately to Fahrenheit, multiply by 9/5.
**Astronomers regard elements heavier than helium as metals.

THE HERTZSRPUNG-RUSSEL DIAGRAM

A H-R diagram has:

- Luminosity on the y-axis using a log scale, or absolute magnitude (going from high to low)
- Temperature (going from high to low) or spectral classes (going from O to M) or colour index (effectively going from blue to red) on the x-axis

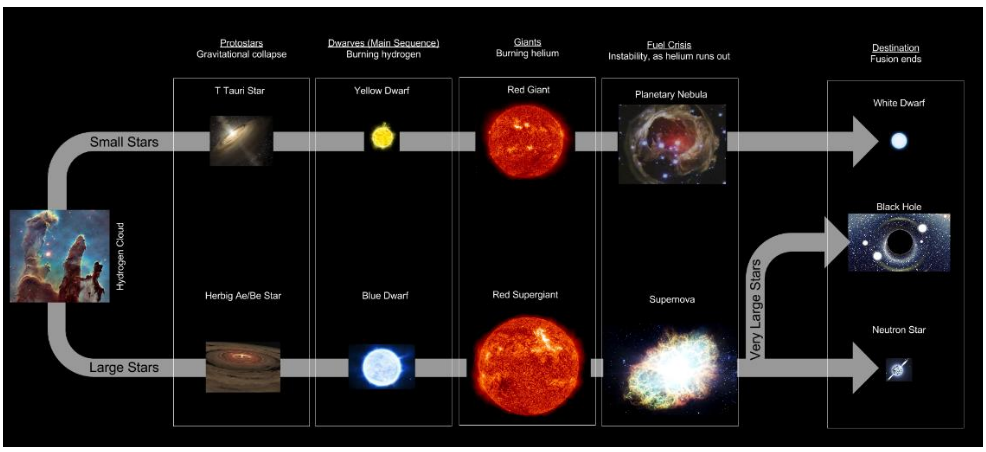


Hertzsprung-Russell Diagram

STELLAR EVOLUTION

A star has only one primary property: its **mass**. All stars begin life on the Main Sequence

- Their mass will determine their position on the Main Sequence, their evolution, the processes that occur inside the star, its lifespan and death.



STAR FORMATION

Stars form from **giant molecular clouds (GMC)** which are relatively **dense** clouds of gas.

- If GMCs contain enough material at a low enough temperature **gravitational collapse and accretion can take place.**
- As the GMC collapses, the potential energy of the molecules is converted to kinetic energy and **temperature rises**. The cloud heads up and begins radiating (emitting black body radiation). It is now a **protostar**.
- Due to their size, protostars can have high luminosities.
- Further collapse leads to further increases in temperature and a reduction in size.
- Once a sufficiently high temperature is reached in the core of the protostar, protons have sufficient kinetic energy to overcome their electrostatic repulsion and **fusion reactions commence**.
- Once fusion commences, the protostar "switches on" and becomes a star.

The new star will now be under the influence of two forces:

- **Gravity** will cause the star to continue to collapse
- **Radiation pressure**, resulting from the energy released by fusion will attempt to make the star **expand**.

The star will initially continue to collapse as gravity will be stronger than the radiation pressure.

- Further collapse will **increase temperature**, which will **increase the rate of fusion** and hence will increase radiation pressure
- Eventually the **radiation pressure** and **gravity will be balanced,** and the star will achieve **hydrostatic equilibrium**.
- The star is now a **main sequence star**.

SUMMARY

1) Cold giant molecular cloud (GMC)
2) Gravitational collapse and accretion
3) Potential energy converted to kinetic energy
4) Temperature increases and radiation begins to be emitted
5) Protostar
6) Further collapse leads to higher temperature
7) Core temperature increases following hydrogen fusion
8) Star (not on main sequence yet)
9) Collapse continues. Fusion releases energy resulting in radiation pressure
10) Radiation pressure increases and balances out gravitational forces
11) Hydrostatic equilibrium reached
12) Main sequence star

MAIN SEQUENCE STARS

Main sequence stars are defined by hydrostatic equilibrium between inward gravitational collapse due to mass, and outward radiation pressure due to fusion.

The mass of the star determines where the on the main sequence the star sits:

- **High mass** stars will be **larger** and will achieve **higher temperatures** as they collapse, meaning the rate of fusion is higher, leading to **higher luminosity**.
- **Low mass** stars will be **smaller**, and will achieve **lower temperatures**, meaning the rate of fusion is slower, leading to **lower luminosity**.

Two possible nucleosynthesis processes occur in main sequence stars, depending on their mass, both fuse hydrogen into helium:

- The proton-proton chain
- The CNO cycle

THE PROTON-PROTON CHAIN

This fusion process occurs in main sequence stars of mass up to 1.3 solar masses.

1) Proton fusion
2) Deuterium formation
3) Deuterium-proton fusion
4) Helium-3 fusion
5) Helium-4 formation

The net result of the fusion reactions occurring in the proton-proton chain is to fuse 4 protons into a helium-4 nucleus, releasing energy and other particles in the process. Beta and gamma decay occurs as well during he process.

THE CNO CYCLE

This fusion process occurs in main sequence stars with **mass of 1.3 solar masses and above**.

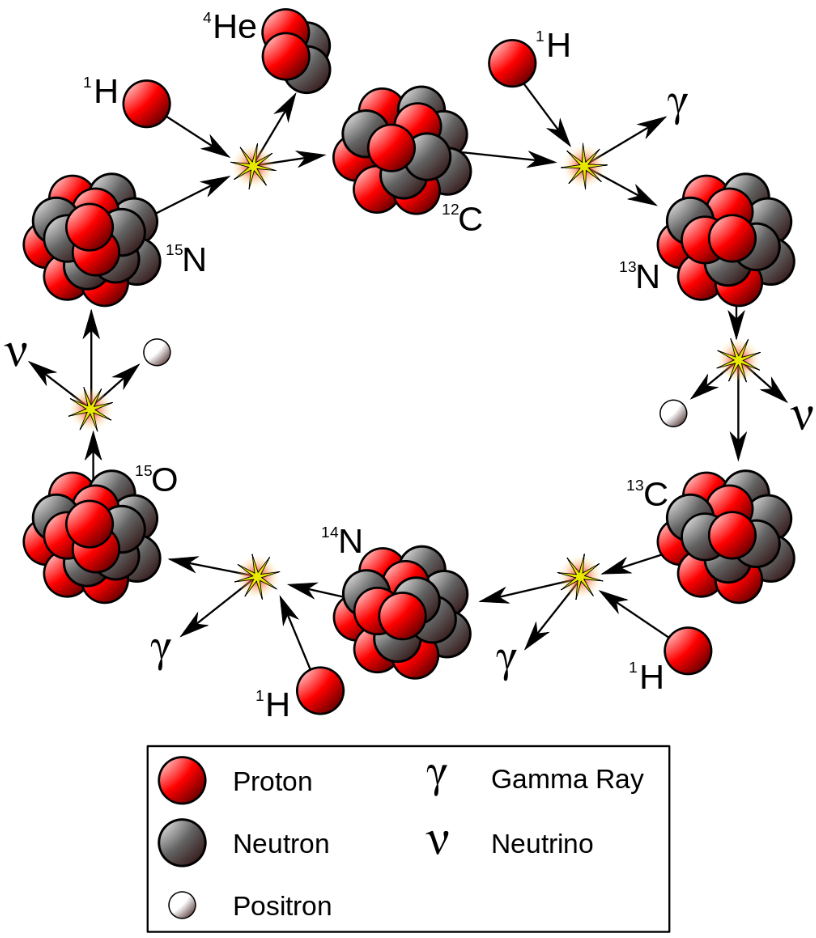
- CNO stands for carbon, nitrogen and oxygen. Various isotopes of these nuclei are involved in this cyclical process.

In the CNO cycle C-12 nuclei acts as a **nuclear catalyst**. They facilitate the reaction but are not changed by it.

The **net result of the CNO cycle is the same as the proton-proton chain**: **four hydrogen nuclei** are progressively fused into a single **helium-4 nucleus**, with gamma and beta decay occurring in the process.

The triple alpha particle in red super giants/giants form carbon.

The CNO cycle dominates in higher mass main sequence stars as more massive stars have hotter cores and can therefore fuse with larger nuclei despite their stronger electrostatic repulsion.



POST MAIN SEQUENCE

Once a main sequence star runs out of hydrogen in its core, hydrogen fusion will cease.

- Fusion ceasing reduces radiation pressure
- Radiation pressure no longer balances gravity, causing the star to **lose hydrostatic equilibrium**.
- The star becomes **unbalanced** and will change.

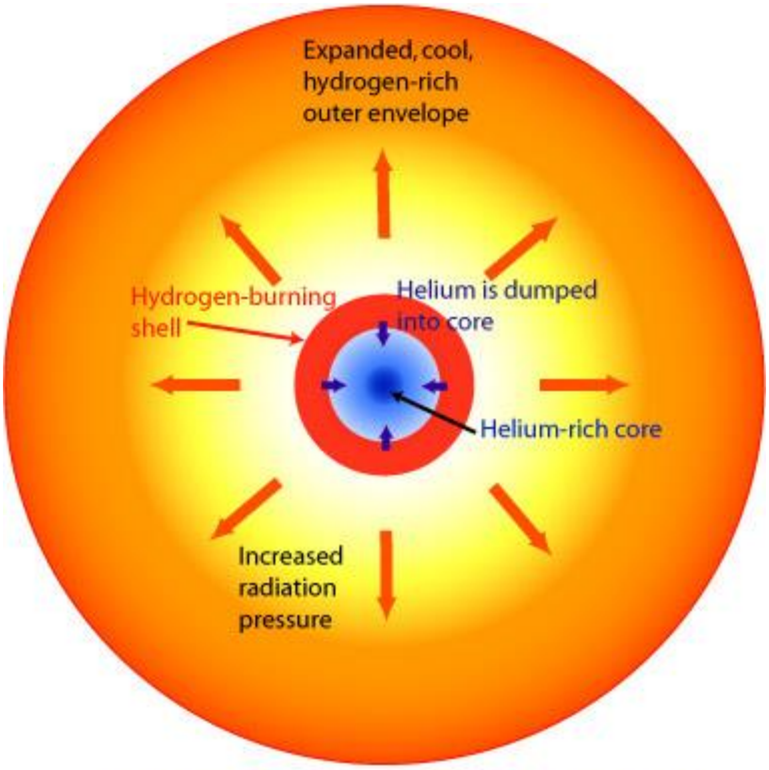
The amount of time it takes for a star to burn through its hydrogen in their core **depends on their mass**.

- Small stars achieve **lower temperatures** and burn hydrogen at a **slow rate**, meaning they will stay on the main sequence **longer**.
- Large stars achieve **higher temperatures** and burn hydrogen at a **faster rate**, meaning they will stay on the main sequence for a **shorter time**.

POST MAIN SEQUENCE (1-3 SOLAR MASSES)

- 1-3 solar masses stars will become **red giants**
- When the hydrogen fusion creases, **outward radiation decreases**. Consequently, layers around the **core collapse inwards**. The core begins to collapse.
  - The collapse of the helium core releases gravitational potential energy as heat. This raises the temperature of the core again.
  - **Hydrogen shell fusion commences**. The helium core is **not performing fusion**.

- Hydrogen shell fusion (shell-burning) continues to produce helium. This helium is deposited into the core, since heavier atoms will sink toward the core (similar to density of objects).



Hydrogen Shell Burning on the Red Giant Branch

- **Expansion** of the outer layer causes **cooling**. Thus, the surface temperature of the star **decreases** and the **spectral class changes**. The star **becomes redder**.
    - Expansion creases once **hydrostatic equilibrium** is **re-established.** The star is now a red-giant
- Hydrogen shell fusion will continue to dump helium into the helium-rich core, causing it to grow and increase in temperature.

Eventually the **temperature of the core** will be **sufficiently high enough** to **ignite helium fusion**. This process produces carbon. The carbon is dumped into the core.

- Helium fusion requires a higher temperature than hydrogen fusion due to significantly higher electric repulsion but **releases more energy than hydrogen fusion**.
- The core expands, **decreasing the rate of hydrogen fusion**.
- Consequently, the increased energy output from helium fusion is somewhat balanced by the decrease output from hydrogen shell burning.
- The outer layers of the star remain relatively unchanged, by the **star continues to be a red giant**, but **produces carbon** in the core.

After the majority of the helium core has been fused into carbon, fusion will cease overall. The star is not massive enough to ignite carbon fusion. The core will collapse into a **white dwarf** and the outer layers will expand into a **planetary nebula**.

WHITE DWARF

White dwarf are the remains of stars cores that **used to be red giant.**

- They are **small** because they consist of only the collapsed core.
- They are **hot** because the red giant's core temperature was initially very high.
- Small size results in **low luminosity**, despite the higher temperature. (Stefan-Boltzmann law $L = \sigma A T^4$)
- White dwarves eventually fade into **black dwarves**

POST MAIN SEQUENCE (3-10 SOLAR MASSES)

Slightly more massive stars will evolve in a similar way like the Sun with **two key differences**:

- They can achieve a **higher temperature** in their core and can fuse **heavier elements**
- This will result in **higher radiation pressure**. And as a result, the star will be **larger** and more **luminous**.

1) Hydrogen fusion in the core produces **helium.** Hydrogen runs out and core collapses and heats up.
2) Hydrogen shell burning, followed by **helium fusion** in the core producing **carbon**. Helium runs out, core collapses and heats up.
3) Hydrogen & helium shell burning. **Carbon fusion** in the core producing **oxygen**. Carbon runs out, core collapse and heats up.
4) Hydrogen, helium & carbon shell burning. **Oxygen fusion** in the core producing **neon, sodium and magnesium**.

POST-MAIN SEQUENCE (>10 SOLAR MASSES)

More massive stars will become **supergiants**. There are two main fusion mechanisms that can occur in larger stars:

1) Nuclei can simply fuse to form a larger nucleus. For example, oxygen fuses to form silicon.
2) Fusion involving **photodisintegration**. Once the core temperature reaches $10^9 K$, **gamma radiation** in the core can lead to **fission**.
   a. A gamma photon is absorbed by a silicon nucleus causing it to photodisintegrate into helium and magnesium

$$^{28}_{14}Si + \gamma \rightarrow ^{24}_{12}Mg + ^{4}_{2}He$$

   b. Helium produced from photodisintegration can fuse with silicon to sulfur

$$^{4}_{2}He + ^{28}_{14}Si \rightarrow ^{32}_{16}S$$

The core will run out of fuel, collapse and increase in temperature.

- Higher temperatures will fuse even heavier elements, releasing more energy
- Higher temperatures will result in higher rates of fusion
- Lighter elements will fuse in shells around the core.

The main fusion steps for a supergiant are as follows:

$$H \rightarrow He \rightarrow C \rightarrow Ne \rightarrow O \rightarrow Si \rightarrow Fe$$

Core fusion will stop when iron is formed as fusion produce nuclei with less binding energy per nucleon. This process absorbs energy and therefore cannot generate any outward pressure to prevent the star's gravitational collapse.

Iron also has the most stable nucleus. As such, the iron cannot undergo fusion without absorbing energy. When too much iron is produced in the core, core fusion ceases and can never restart.

STAR DEATH

When fusion in the core creases, the star dies.

- The core of the star collapses
- The outer layers of the star are ejected

For **lower mass stars**, the core **collapses** to a **white dwarf** and the outer layers form a **planetary nebula**. The planetary nebula forms as residual radiation pressure causes the outer layers to keep expanding.

Stars with mass around 8 solar masses or lower will die this way, resulting in white dwarfs between 0.6 to 1.4 solar masses.

Larger stars will explode as a **supernova**. The outer layers are ejected as a **supernova remnant** and the core collapses to form a **neutron star** or a **black hole**. Core collapse once fusion creases is very rapid, resulting in a shockwave and a huge explosion.

- The outer layers continue expanding uncontrollably into the surrounding space forming the **supernova remnant**.

Supernovae are extremely energetic events and allow the fusion of heavier elements to occur, forming elements heavier than iron. Massive stars exert large enough gravitational forces on their cores during a supernova to bring iron nuclei close enough to fuse. (I.e. they can convert enough gravitational energy to kinetic energy to fuse the iron nuclei).

The core of star collapses into one of two options:

- 8-25 solar masses: **neutron stars**
- >25 solar masses: **black holes**

NEUTRON STARS

Neutron stars have extremely high densities. High pressure causes nuclear reactions where electrons and protons combine to form neutrons.

- Neutron stars are almost entirely composed of neutrons with a density similar to that of atomic nuclei.
- The neutron star will be around 1.4-3.2 solar masses
- Their radius will be relatively small, around 30km
- Their temperature will be million of degrees, hence they emit radiation in the x-ray part of the spectrum and virtually no visible light.

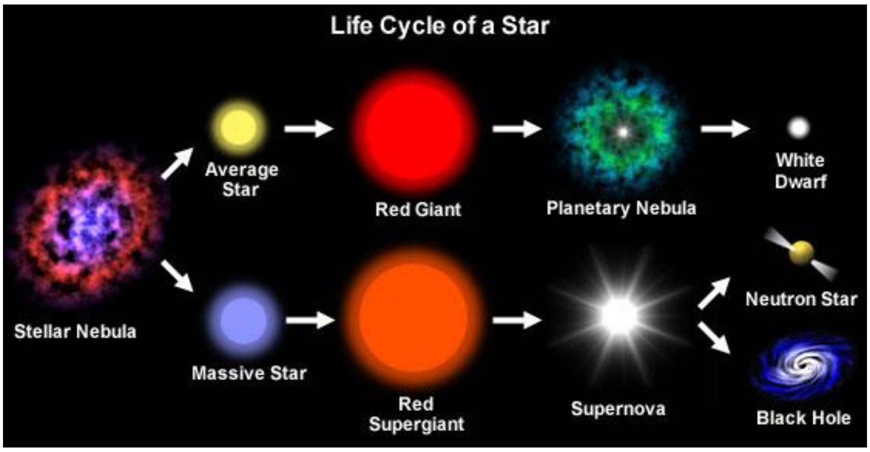
BLACK HOLES

Black holes form when the mass of stars is so high that the **collapsing core forms a singularity** and the **escape velocity** of the collapsing core exceeds the speed of light.

Black holes do not have a solid boundary. The point of no return around the singularity is the position in space where the escape velocity reaches the speed of light, and is called the **event horizon**.

Black holes slowly radiate away its mass as **light** in a quantum event horizon effect called **Hawking Radiation**.

SUMMARY



SUMMARY: NUCLEOSYNTEHSIS

Atomic Number/Element	Origin	Process
1 Hydrogen	Everywhere	Formed when energy was converted to matter after the Big Bang, eventually forming protons
2 Helium	Everywhere, and inside all stars	Formed during the period of nucleosynthesis after the Big Bang where protons and neutrons could fuse to form helium Formed inside all stars as a result of hydrogen fusion
3-6 Lithium to Carbon	Inside Red Giants and Supergiants	Formed by fusion reactions of helium and heavier elements (Small amounts of lighter elements also formed after the Big Bang during the period of nucleosynthesis)
7-26 Nitrogen to Iron	Inside Supergiants	Formed by fusion reactions of heavier elements. Can involve photodisintegration reactions
>26	In Supernovae	Formed by fusion reactions of heavier elements that absorb energy and hence are only possible in environments where huge amounts of energy are released.

PHYSICS

Working Scientifically

RELIABILITY, VALIDITY AND ACCURACY

RELIABILITY

WHAT IS IT?

Reliability is the extent to which the experiment **yields the same result each time**.

- A set of results can be reliable but inaccurate if the same incorrect result is obtained in every trail.
- Note that simply repeating an experiment will not improve reliability, but it can allow the experimenter to determine if it is reliable.
- **Random errors** caused by unknown and unpredictable changes in the experiment reduces reliability.

HOW TO IMPROVE RELIABILITY

A measurement is reliable if you **repeat it** and get the **same or similar answer** over and over again, and an experiment is reliable if it **gives the same result** when you **repeat the entire experiment**.

HOW DO YOU TEST RELIABILITY?

Reliability can be tested through **repetition**. The more similar repeated measurements are, the more reliable the results. However repetition alone doesn't make measurements reliable, it **allows you to check whether or not they are reliable**.

Improving reliability is a different matter to testing it. **The reliability of single measurements is not improved through repetition**, but through the **design of the experiment**. Implementing a method that reduces **random errors** will improve reliability. However, the entire result of the experiment can be improved through **repetition** and **analysis**, as this may reduce the effect of random errors.

For example, in an experiment to determine the change in temperature when ammonium nitrate $(NH_4)(NO_3)$ is dissolved in water:

- Using a fairly large mass of ammonium nitrate ensures that weighing errors are minimised and that a large change in temperature is observed (also minimising error).
- Repeating the same experiment multiple times, removing outliers and averaging the results will improve reliability.

If the data are close to the correct trend line, it can be determined that the results are reliable.

VALIDITY

Validity is the extent to which the experiment **tests the hypothesis** or **addresses the aim**.

- In other words, are you actually measuring what you're trying to measure?
- Validity depends on how appropriate the experiment procedure is e.g. whether a suitable control was used.

Several aspects of the experiment can contribute to validity: the **equipment**, the **experimental method**, and the **analysis of the results**.

- The equipment must be **suitable** for carrying out the experiment and taking **necessary measurements**.
- Control variables.
- Independent (x), dependent (y)

HOW TO IMRPOVE RELIABILITY

The method (including the analysis) may contain some assumptions that need to be satisfied. The experimental method must ensure that **all assumptions are satisfied**, otherwise, you end up using a method or analysis that is inappropriate and the result will be invalid.

Techniques used to improve reliability of results in the ammonium nitrate dissolution experiment described above depends on the hypothesis.

- If the hypothesis is: "Ammonium nitrate produces temperature changes when dissolved in water", and there is a recorded temperature change, the experiment is valid.
- If the hypothesis is: "The heat dissolution of ammonium nitrate is +2567 kJ/mol" and your experiment has significant heat loss, the experiment is invalid.

ACCURACY

Accuracy is the extent to which the calculate value **differs from the true, accepted value**.

HOW TO IMPROVE ACCURACY

The accuracy can be improved through the experimental method if each **single measurement is made more accurate**, eg through the choice of equipment. Implementing a method that **reduces systematic errors** improves accuracy.

Techniques used to improve the accuracy of results in determining the heat of dissolution of ammonium nitrate experiment are:

- Ensure the equipment is capable of determining the heat of solution i.e. Adequate thermal insulation, appropriate quantity of salt, etc.

HOW TO TEST ACCURACY

- Comparing measurements to value expected from theory for single measurement
- Comparing final experimental result to accepted value for entire experiment's result

TYPES OF ERROR

Error is a complementary concept. A measurement with high accuracy will have low error. A measurement with large error is inaccurate.

MISTAKES

Mistakes are avoidable errors. A measurement that involves a mistake must be rejected and not included in any calculations or averaged with other measurements of the same quantity.

SYSTEMATIC ERROR

Systematic errors are errors that are **consistent** and will **occur again** if the investigation is repeated in the same way. They are usually a result of instruments that are not calibrated correctly or methods that are flawed.

Systematic errors will **shift measurements** from their **true value**, by the same amount or fraction in the same direction all the time. This usually arises from **problematic** or **incorrectly used equipment** eg. Poor calibration.

RANDOM ERRORS

Random errors occur in an **unpredictable manner** and are generally small.

Random errors will **shift each measurement from its true value** by a **random amount** in a **random direction**.

TYPES OF ERRORS

Error	Description	Systematic or Random Error
Scale Error	If a piece of equipment is not calibrated correctly, all measurements will be offset by the same fraction.	Systematic error
Zero Error	If a piece of equipment has an offset that is not zero when there is nothing on it, all measurements will be offset by the same amount.	Systematic error
Parallax error	If you make a measurement by comparing an indicator against a scale, the angle at which you view it will affect the reading.	Systematic error if you always view it from the same angle. Random error if you view it from a random angle at each time.
Errors arising from the environment	Ideally, the control variables are kept constant, but some are beyond your control.	Changes to the control variables result in both systematic and random errors. One consistent change will give a systematic error. Random changes will give random errors.
Measurement error from insufficient precision	If you're measuring something that falls between two markings on a scale, you cannot measure its precise value and will need to round up or down.	Random error

UNCERTAINTY

UNCERTAINTY

When there is a range of measurements of a particular value, the mean must be accompanied by the uncertainty for your results to be presented as a mean in an accurate way.

Uncertainty is calculated by:

Uncertainty $= \pm(\text{maximum difference from the mean})$

ABSOLUTE UNCERTAINTY

The absolute uncertainty is related to the **precision of the instrument** being used. Absolute uncertainty is **equal to half the smallest unit of measurement**.

$$\text{Absolute Uncertainty} = \frac{1}{2} \times \text{precision}$$

PERCENTAGE UNCERTAINTY

Percentage uncertainty, also known as relative uncertainty, is another way of stating how **precise a measurement is**.

$$\text{Percentage Uncertainty} = \frac{\text{absolute uncertainty}}{\text{measured value}} \times 100$$

UNCERTAINTIES

ABSOULTE UNCERTAINTY

- **The limit of reading** is the smallest unit into which a measuring scale is made. When a measurement is made, any claim for a precision better than the limit of reading is an estimate only.
- **Absolute uncertainty (absolute error)** is the range above and below the quoted value of the measurement, within which we would expect repeated measurements to be found.
    - Absolute error is often written as Δx if the measurement is x
    - $\text{Absolute Uncertainty} = \frac{1}{2} \times \text{limit of reading}$
- For analogue scales (such as clock face or a ruler), the AE is often quoted as being **half the limit of reading.** Thus, on a typical school ruler this AE would be 0.5 mm.
- For digital scales, the AE is often quoted as being equal to the limit of reading (ie. we take a more conservative approach).
- **Relative error (fractional error):** is a measure of how important the error is. For example, being out by $\pm 1\ cm$ is a big issue if measuring the thickness of a book, but not so important if measuring the distance from home to school. Relative error can be converted to a percentage:
    - **Fractional uncertainty** is found by using: $\frac{absolute\ uncertainty}{absolute\ measurement} = \frac{\Delta x}{x}$
    - **Percentage uncertainty** is found by using: $\frac{absolute\ uncertainty}{absolute\ measurement} \times 100$
        - $\Delta_{\%} x = \frac{\Delta x}{x} \times 100$

ERRORS IN REPLICATE MEASUREMENTS

There are **several accepted conventions** used in the physics community.

Suppose a series of time measurements are taken for a vibrating spring:

- The error in the average time will be $\frac{range}{2}$
- Pearson Textbook however uses **maximum difference** between the mean and any one piece of data used in the calculation. ('Residual' technique where the error in the mean time is taken to be the largest deviation from the mean of any one of the measurements)
- Uncertainty $= \pm(\text{maximum difference from the mean})$

UNCERTAINTIES IN ADDITION AND SUBTRACTION

If $y = a \pm b$, then $\Delta y = \Delta a + \Delta b$ (Add the error)

ERRORS WHEN MULTIPLYING AND DIVIDING

Add individual fraction or percentage errors.

If $y = \frac{ab}{c}$, then $\frac{\Delta}{y} = \frac{\Delta}{a} + \frac{\Delta}{b} + \frac{\Delta}{c}$

$L_1 = 10.25\ \pm 0.05cm, L_2 = 15.45\ \pm 0.05cm. L_1 \times L_2$

$$L_1 = 10.25\ \pm 0.05cm\ \%unc = \frac{0.05}{10.25} \times 100 = 0.5\%$$

$$L_2 = 15.45\ \pm 0.05cm\ \%unc = \frac{0.05}{15.45} \times 100 = 0.3\%$$

$L_1 \times L_2 = 158.36\ \%uncertainty = 0.5 + 0.3 = 0.8\%$

RAISED TO THE POWER OF

If $y = a^n$, then $\frac{\Delta y}{y} = \left| n\left(\frac{\Delta a}{a}\right) \right|$

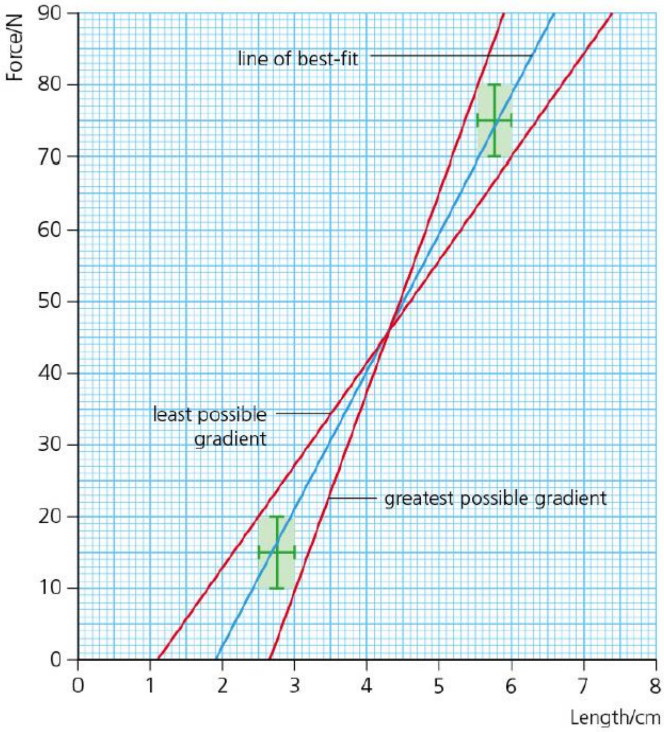
UNCERTAINTIES IN GRAPH

Error bars are used for each plotted graph. However if the error is negligible then it may be justified in ignoring it and just plotting the error bar for one of the two variables.

Any legitimate line of best fit will pass through the error bar range. (Excel or Logger Pro can be used)

When calculating a gradient from a graph, it is important to estimate the magnitude of uncertainty in it. This can be done by drawing onto the graph two lines of "worst fit" (AKA lines of minimum and maximum gradient)

These are down by imagining a square drawn around the error bars of the two extreme data points.



$$\Delta x = \frac{x_{max} - x_{min}}{2}$$

$$\Delta m = \frac{m_{max} - m_{min}}{2}$$

$$\Delta Intercept = \frac{I_{max} - I_{min}}{2}$$